

Exponential Parity Correlation Bounds for Depth-Three QAC Circuits

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Abstract

We study one-bit parity computation by quantum circuits with three layers of disjoint generalized Toffoli gates and arbitrary one-qubit unitaries. Ancillary width and gate count may be any finite numbers; ancillas start in an input-independent product state, and final garbage is unrestricted. The principal bound developed here is $|\beta(C)| \leq 6K_0 e^{-\kappa n}$ on uniform-input parity correlation, for absolute $K_0, \kappa > 0$ independent of both size and width. It rules out any fixed bounded error for sufficiently many inputs, whatever the finite circuit size. The proof first establishes a shorter $K \exp(-cn^{1/3})$ bound by robust restriction and signed-source retention. Exact heavy-gate expansion and a rank-two first-layer correction then give the exponential rate.

1 Introduction

Parity is a basic test of the ability of a shallow circuit to combine information from all of its inputs. For $n \geq 1$, it is the Boolean function $\text{PARITY}_n(x) = x_1 \oplus \cdots \oplus x_n$, where $\oplus$ denotes addition modulo two. Its value changes whenever any one input bit is flipped, yet a tree of CNOT gates computes it in $\lceil \log_2 n \rceil$ layers. The question for QAC⁰ is whether a polynomial-size family of circuits with unbounded-arity generalized Toffoli gates and arbitrary one-qubit unitaries can compute parity in constant depth. A generalized Toffoli flips one target qubit precisely when all its control bits are one; a CNOT has one control. Moore’s reductions between coherent parity, quantum fanout, and cat-state preparation explain the central role of this question [3]. Here we study its depth-three, single-output version and develop an exponential upper bound on the parity correlation of every circuit in that model.

1.1 Motivation, model, and overview of the result

The classical benchmark is the exclusion of parity from polynomial-size AC⁰. Håstad’s switching-lemma method gives strong size lower bounds, while the Linial–Mansour–Nisan theorem describes the concentration of low-depth Boolean functions at low Fourier degrees [1, 2]. Transferring these methods to quantum circuits is difficult: quantum gates can create interference and entanglement, while the unbounded support of a Toffoli gate prevents a direct application of the usual bounded-fan-in light-cone argument. Moreover, the classical convention of free fanout has no counterpart in the model considered here.

Depth throughout this manuscript counts layers of multiqubit gates; arbitrary one-qubit unitaries can appear at the layer boundaries. Within each layer the gate supports are disjoint, including

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all control wires. We allow at most three such layers. A circuit receives one computational-basis copy of each input bit and any finite number of ancillas in an input-independent product state, possibly mixed. The circuit and its final measurement are fixed independently of the input. Only the measured answer on one designated qubit is specified: the inputs may be overwritten and all other outputs may contain garbage. Section 2 gives the formal model and includes general binary measurements on that qubit.

For a circuit C , let $a_C(x)$ be the probability of output zero minus the probability of output one, and write

$$\beta(C) = \mathbb{E}_{x \in \{0,1\}^n} (-1)^{x_1 + \dots + x_n} a_C(x).$$

The expectation is uniform over all 2^n inputs. If X is a uniform input and Y is the measured bit produced by C on X , then

$$\mathbb{P}[Y = \text{PARITY}_n(X)] = \frac{1 + \beta(C)}{2}.$$

Thus the uniform-average advantage over random guessing is $\beta(C)/2$; allowing an output-bit flip makes it $|\beta(C)|/2$. Our principal result, Theorem 2.1, is

$$|\beta(C)| \leq 6K_0 e^{-\kappa n} \quad \text{for absolute } K_0 \geq 1, \kappa > 0.$$

The bound holds for every $n \geq 1$, and the constants are independent of gate count, ancillary width, gate arity, local bases, and the chosen output measurement. Consequently, for every fixed $0 \leq \varepsilon < 1/2$ there is $n_0(\varepsilon)$ beyond which no such circuit has uniform-average error at most ε . This also rules out worst-case error at most ε , since a worst-case guarantee implies the same average guarantee. More generally, no circuit family in this model has positive uniform-average advantage $\exp(-o(n))$. The strongest explicit specialization developed in the appendices is $|\beta(C)| \leq 1800 \exp(-2^{-27}n)$ (Corollary D.2); its constants are conservative, and for small n the elementary bound $|\beta(C)| \leq 1$ is stronger. All logarithms without a subscript are natural.

The distinction between a size lower bound and a size-independent obstruction is essential. Although QAC^0 ordinarily refers to polynomial-size families, the bound above holds for every finite three-layer circuit, including exponentially large circuits. It does not settle parity for arbitrary constant depth. It also does not grant free fanout, multiple supplied copies of the input, entangled initial advice, intermediate postselection, or global classical processing of the output. These changes to the model can alter its computational power.

1.2 Past work and related results

Fanout and the role of the output specification. Coherent parity preserves a data register x and performs the target update $b \mapsto b \oplus \text{PARITY}_m(x)$ unitarily. Fanout instead maps $|b, x_1, \dots, x_m\rangle$ to $|b, x_1 \oplus b, \dots, x_m \oplus b\rangle$. Conjugating by a Hadamard on every wire interchanges these operations; counting and modular gates are also connected to them by constant-depth reductions [3, 4]. Høyer and Špalek [5] demonstrate the computational power available when fanout is supplied as a primitive. Takahashi and Tani [8] show that, when fanout is available, bounded-fan-in, unbounded-Toffoli, and threshold gate sets have the same exact constant-depth computational power. Grier and Morris [16] obtain approximate fanout from quantum threshold gates without assuming fanout as a primitive. These connections motivate parity lower bounds, but reductions between gate synthesis, state preparation, and one-bit decision problems need not preserve an exact depth bound or a measurement convention. A *clean* coherent implementation also restores its work ancillas to their initial state. A coherent, input-preserving implementation with garbage still places more conditions on the output than the single measured bit required here.

Lower bounds with restricted ancillary space. Fang, Fenner, Green, Homer, and Zhang [6] prove a logarithmic depth lower bound for clean parity computation with constantly many ancillas, together with a depth–ancilla tradeoff. Bera [7] gives another method excluding exact parity in constant depth without ancillas. These results address depths beyond three, but their ancillary restrictions prevent them from treating the arbitrarily wide circuits considered here.

In the spectral results below, an operator has *Pauli degree at most k* if its tensor-Pauli expansion has no term with more than k nonidentity factors. The *approximate degree* of a Boolean function is the least degree of a real polynomial that approximates it within error $1/3$ on every Boolean input.

Nadimpalli, Parham, Vasconcelos, and Yuen [11] introduce Pauli-spectrum methods and prove low-degree concentration with limited ancillas. For each fixed depth d , their Theorem 33 implies that a polynomial-size, depth- d circuit with at most $\frac{1}{2}n^{1/d}$ ancillas has uniform-input parity success at most $1/2 + 2^{-\Omega(n^{1/d})}$ as $n \rightarrow \infty$.

Anshu, Dong, Ou, and Yao [13] obtain spectral-norm low-degree approximations and the ancillary lower bound $\tilde{\Omega}(n^{1+3^{-d}})$ for bounded-error computation of functions of approximate degree $\Theta(n)$. Dong, Ou, and Yao [14, Theorem 3.1 and Remark 3.2] subsequently improve the degree exponent from $1 - 3^{-d}$ to $1 - 2^{-d}$ for inverse-polynomial spectral-norm error, obtaining the stronger parity ancillary bound $\tilde{\Omega}(n^{1+2^{-d}})$. Their general approximation theorem has a restricted error-parameter range, so this is not a uniform improvement in every error regime. Here tildes suppress logarithmic factors. These structural results cover arbitrary constant depth and yield applications beyond parity. Our theorem addresses only three layers but allows every finite ancillary width with product initialization. The spectral-norm bounds of [13, 14] also permit entangled initial ancillas; our result removes the restriction on their number, not the product-state hypothesis of our model.

Unrestricted-width results at depths two and three. Padé, Fenner, Grier, and Thierauf [9] rule out clean simulation of the quantum parity gate on more than three qubits in depth two. Fenner, Grier, Padé, and Thierauf [15] strengthen this to a sharp exact obstruction for parity of more than three inputs on a separately initialized answer qubit, allowing garbage and arbitrarily many product-initialized ancillas [15, Definition 2.7 and Theorem 6.1]. Their Remark 6.10 permits certain more general initial states subject to separability conditions; it does not allow arbitrary entangled ancillas. The earlier threshold counts total parity-gate arity, whereas the later threshold counts data inputs excluding the initialized answer. Neither should be transferred unchanged to our convention, which also permits measuring an input wire.

Rosenthal [10] proves an unrestricted-width depth-two fidelity bound of $1/2 + \exp(-\Omega(n))$ for coherent parity and fanout on appropriate quantum test inputs, even allowing garbage. Here fidelity means squared overlap of pure states. This does not by itself bound the one-bit success measured by $\beta(C)$. For the same coherent-fidelity tasks, his Corollary 1.6 gives an $\Omega(n/d)$ multiqubit-gate lower bound for fidelity advantage $\exp(-o(n/d))$ above $1/2$. These are asymptotic approximation statements rather than sharp finite-input exact thresholds.

The closest predecessor is Joshi, Tal, Vasconcelos, and Wright [17]. Their Theorem 1.2 excludes exact single-output parity for $n > 100$, regardless of circuit size or product-ancilla width. At depth two they prove parity correlation $\exp(-\Omega(\sqrt{n}))$ and total influence $O(\log n)$ with unlimited ancillas and gates. The exponential depth-three correlation bound here strengthens that asymptotic exact obstruction to average-case approximation and also improves the depth-two parity rate as a special case. It does not improve their finite-input exact threshold, nor does it replace their broader influence bounds or their results for majority.

Upper bounds and the limits of a fixed-depth theorem. Rosenthal [10] constructs approximate parity in depth seven and, more generally, in depth $d \geq 7$ with size $\exp(\text{poly}(n^{1/d}) \log(n/\varepsilon))$ for error ε in phase-dependent fidelity on every quantum input, via approximate nekomata preparation. In his convention the phase-dependent fidelity of unit vectors ψ, ϕ is $1 - \|\psi - \phi\|_2^2$; an m -nekomata is a state $(|0^m\rangle|\psi_0\rangle + |1^m\rangle|\psi_1\rangle)/\sqrt{2}$ with normalized, not necessarily orthogonal, ancillary vectors ψ_0, ψ_1 . Unlike a cat state tensored with fixed garbage, its two branches may have different ancillary states. Grier, Morris, and Wu [18] subsequently obtain exact parity in constant depth with exponential size using amplitude amplification. They also show that polynomially many supplied input copies enable bounded-error TC^0 computations by QAC^0 . These constructions explain both the depth restriction and the single-copy assumption in our theorem. Approximate preparation of a large nekomata does not by itself constitute a three-layer circuit whose one measured qubit computes parity.

Recent structural characterizations and neighboring models. Gretta, Gupta, and Joshi [19] connect parity lower bounds to Fourier concentration of QAC^0 and relate inverse-polynomial average parity correlation to exact fanout through QAC^0 reductions. Joshi et al. [17] likewise discuss conversion from approximate to exact parity. Such reductions allow additional constant depth, so an exact depth-three exclusion alone does not give the approximation theorem at the same depth. Our proof bounds $\beta(C)$ directly.

Xu and Li [21] characterize symmetric Boolean functions in terms of the fanout size needed to compute them; their equivalence uses worst-case approximation with inverse-polylogarithmic advantage. This places parity within a broader family of fanout-related problems, without supplying the fixed-depth average-case estimate proved here. For geometrically restricted circuits, Dong, Ou, and Yao [20] obtain nearly logarithmic depth lower bounds for parity in one dimension with unlimited ancillas, and stronger bounds for contiguous inputs; their ancillas may start in arbitrary states. Their spatial constraint differs from our unrestricted gate geometry. Related lower bounds against classical AC^0 with simple quantum preprocessing [12] concern a different composition of quantum and classical computation.

1.3 What is novel

The new claim is the combination of depth three, a single measured output, unrestricted finite size and product-ancilla width, and exponential decay of uniform-input parity correlation. The previous paragraphs separate results that have some, but not all, of these features. The shorter $K \exp(-cn^{1/3})$ argument already rules out fixed bounded error. The appendices strengthen its rate to $\exp(-\Omega(n))$.

For clarity, we introduce the terminology used in describing these contributions. First reduce mixed product ancillas to pure product states by convexity, as justified in Section 2. A *one-input source* is a first-layer component containing one variable input and otherwise fixed product-state qubits; an untouched input is a singleton source. Write its two output states as ρ_0, ρ_1 , its signed operator as $F = (\rho_0 - \rho_1)/2$, and its average state as $\bar{\rho} = (\rho_0 + \rho_1)/2$. A *port* is a block in a partition of the source wires into nonempty, pairwise disjoint sets. In the network application, it is the intersection with one second-layer gate, or the block of idle source wires. The *owner* port contains the variable input wire. Marginals such as $\bar{\rho}_B$ are obtained by tracing out all wires outside the indicated union of ports B . For $B = \emptyset$, the marginal $\bar{\rho}_B$ is the scalar one. We use $\|X\|_1 = \text{Tr} \sqrt{X^* X}$ for the trace norm and $\lambda_{\max}(\rho)$ for a density operator's largest eigenvalue. The technical contributions are as follows.

- (i) *A uniform budget for one input spread across many ports.* Assign each port the mixedness

weight $h_j = 1 - \lambda_{\max}(\bar{\rho}_j)$. Lemma 4.1 proves

$$h_0 = \frac{1}{2}, \quad \sum_j h_j \leq C * s = \frac{1}{2} + \frac{2}{e},$$

where $C * s$ is an absolute constant and port 0 is the owner. The lemma also gives a nondecreasing majorant of $\|\text{Tr}_{\mathcal{J} \setminus A} F\|_1^2$, where $\mathcal{J}$ is the set of ports and A is the retained subset, with mean at most $3/4$ under independent fair port retention and oscillation at most $4h_j$ when port j is toggled. The budget and the retention estimate hold regardless of the number or dimensions of the ports.

- (ii) *A global bound that tolerates shared second-layer gates.* Theorem 5.1 combines those source estimates across an arbitrarily wide layer. Different sources can share a gate and hence a random retention decision; their retained norms are not independent. The weighted budget controls this dependence through the total change caused by each gate. The exponential refinement (Theorem B.3) bounds the absolute full r -input parity coefficient by $K_0 e^{-ar}$ for an event whose effect is a tensor product of positive contractions over the ports. It bounds the signed average of the event probabilities, not the individual event probabilities. Here $0 \leq E \leq I$ defines a positive contraction, r is the number of sources, and $K_0, a > 0$ are absolute. The auxiliary resource is tensor-independent of the sources but may be internally entangled. In the exact expansion it may even be a nonpositive complex operator of trace norm at most one; in that case the corresponding functional is not itself a probability. Such operators are auxiliary algebraic terms, not initially entangled advice.
- (iii) *Retention and loss after joint rank-one insertion.* Heavy-gate corrections can touch several ports of the same source. Lemmas B.1 and B.2 handle these ports as one inserted register and retain bounds in terms of the original source weights. Appendix C supplies a second route using a centered rank-one inequality for arbitrary Hermitian operators. For a union B of original source ports, Lemma D.1 gives the convex joint-loss estimate

$$1 - \lambda_{\max}(\bar{\rho}_B) \geq \Phi\left(\sum_{j \in B} h_j\right), \quad \Phi(t) = \frac{t}{4 \log(4/t)} \quad (0 < t \leq C * s), \quad \Phi(0) = 0.$$

The argument of Φ belongs to $[0, C * s]$ by the source budget. These ingredients permit summing dense correction sets. Separate marginal penalties within one source cannot be multiplied: for the CNOT source $F = (|00\rangle\langle 00| - |11\rangle\langle 11|)/2$, both one-qubit marginals of $\bar{\rho}$ have largest eigenvalue $1/2$, but $|\text{Tr}(|00\rangle\langle 00|F)| = 1/2 > 1/4$.

- (iv) *An exact treatment of large first-layer components.* After signing and averaging its k input bits, a first-layer reflection changes the signed component operator by a rank-at-most-two correction of trace norm at most $4 \cdot 2^{-k}$. Theorem B.4 retains this correction as one unnormalized resource operator and sums the resulting branches by a generating function. This avoids the loss in the exponent from deleting all large first-layer gates and is the step that restores exponential decay in the original number n of inputs.

The centered inequality and the convex estimate are proved here as parts of the argument; no separate historical priority claim for general matrix inequalities is asserted. For comparison, Proposition 7.1 gives a zero-ancilla construction: two CNOT layers compute parities of disjoint groups of at most four inputs, and one Toffoli combines the $r = \lceil n/4 \rceil$ representatives. For $r \geq 2$ its

correlation, after a possible output NOT, is 2^{2-r} , so its uniform-average advantage is 2^{1-r} . It uses at most $n - r + 1$ multiqubit gates. This establishes the qualitative tightness of exponential decay, not an optimal decay constant or a worst-case success guarantee. In particular, exact parity is possible for $n \leq 8$ within three layers; no sharp depth-three exact-computation threshold is asserted here.

1.4 What is derived from previous work

Several essential ingredients are inherited or adapted rather than claimed as new. We make their provenance explicit.

- (i) *Reflection normalization.* The representation $D = I - 2|v\rangle\langle v|$, with v a product vector, and absorption of one-qubit basis changes follow Rosenthal’s normal form [10, Proposition 2.1]. The equivalence between Toffoli and controlled phase gates is already used in earlier QAC lower bounds [6].
- (ii) *Restriction and source decomposition.* Removing first-layer gates that touch many inputs and keeping one variable per remaining component adapt the cleanup in Joshi–Tal–Vasconcelos–Wright [17, Section 5.1, Claims 5.2–5.3]. Their equations (67)–(79) also use the tensor-product source decomposition and signed matrix Fourier coefficients. Tensorizing these coefficients follows from linearity and is not claimed as new. Lemma 3.1 supplies the multi-gate mean error bound used here. The added estimates are the port budget, retention control, and global summation described above.
- (iii) *The final-output interface.* Pulling a one-qubit observable backward through the final reflection uses its elementary block-diagonal form, as in the reflection calculus of [10, 17]. Lemma 6.1 records the resulting total absolute coefficient mass of at most six. Spectral decomposition and this local algebra are not a separate circuit lower-bound mechanism.
- (iv) *Concentration and norm tools.* The Doob-martingale bounded-differences inequality is standard McDiarmid concentration [22]; its exponential-moment step uses the bounded-variable estimate of Hoeffding [23]. Trace-norm duality, Schatten Hölder, convexity, and Jensen’s inequality are also standard. What is specific here is the source weight that makes their application independent of circuit width. The fair vertex mask is the exact identity $E = (I + (2E - I))/2$ for a positive contraction. It is distinct from the support-based activation predicate in [17, Definition 3.4] and makes no assertion that measured quantum outputs are independent.

1.5 Overview of the proof strategy

The proof isolates parity information at the first layer and follows its signed contribution through product measurement events at the second layer. The final layer is handled locally at the measured qubit. The intermediate product events are algebraic observables, not additional measurements performed by the circuit.

The shorter restriction argument. Fix an integer $k \geq 2$ with $n/(k-1) \geq 1$. Deleting all first-layer gates containing at least k original inputs costs at most $4\sqrt{(n/k)2^{-k}}$ in correlation. In every undeleted input-containing component keep one variable and fix the others; keep all inputs from deleted gates and all untouched inputs as singleton sources. This leaves $r \geq \lceil n/(k-1) \rceil$ sources. The estimate holds for every assignment z to the fixed inputs. Indeed, for the deleted circuit C^- ,

$$\beta(C^-) = \mathbb{E}_z(-1)^{|z|}\beta(C^-|_z),$$

so uniform bounds on the restricted coefficients give the required bound without assuming that a restriction preserves worst-case correctness. For each source, $F_i = (\rho_{i,0} - \rho_{i,1})/2$ is its signed average. Tensoring the F_i extracts the restricted full parity coefficient. Input-free components form a separate resource, which need not factor over its second-layer ports.

Partition each source into ports according to its second-layer gates. For $w \geq 1$, call a gate heavy if its incident weight $\sum_i h_{ig}$ exceeds w , where h_{ig} is the weight of source i 's port at gate g (zero if there is no such port). The source budgets imply at most $C * sr/w$ heavy gates, and deleting them costs at most $8C * s(r/w)e^{-w}$ in the event coefficient. Their base product effects are split into separate port effects. Each remaining gate effect, or separate port effect, is a vertex. Expanding $E_g = (I + (2E_g - I))/2$ introduces an independent fair coin at each vertex to decide which ports to retain. This is an exact algebraic average. The sum F_* of source majorants satisfies $\mathbb{E}F_* \leq 3r/4$ and has squared vertex oscillations summing to at most $16C * srw$. The source majorants may be dependent because different sources can share a vertex. Bounded differences controls F_* ; AM-GM then controls the product of retained source norms. Including the deletion cost and choosing $w = \sqrt{r}$ gives $\exp(-\Omega(\sqrt{r}))$ decay. The final-output expansion has total absolute coefficient mass at most six. Using $r \geq n/(k-1)$ and choosing $k \asymp n^{1/3}$ balances this bound with the first-layer deletion cost and yields $K \exp(-cn^{1/3})$.

The exponential refinement. Now fix a sufficiently large absolute constant w , independent of r . For a heavy second-layer gate $D_g = I - 2\Pi_g$ and its product effect P_g , use the exact identity

$$D_g P_g D_g = P_g - 2\Pi_g P_g - 2P_g \Pi_g + 4\Pi_g P_g \Pi_g.$$

Each of the three correction operators has rank at most one and factors over the incident ports. For a selected set of s correction gates, the sum of absolute coefficients over their types is $(2 + 2 + 4)^s = 8^s$; no additional factor 3^s is needed. For source i , tensor the selected port contractions into a single rank-at-most-one contraction T_i on their union B_i and put $K_i = \text{Tr}_{B_i}[(T_i \otimes I)F_i]$. For $B_i = \emptyset$, use the scalar $T_i = 1$, so $K_i = F_i$. The operators K_i remain unnormalized and may be non-Hermitian.

Post-insertion retention still gives a constant deficit in the mean of each squared-norm majorant, with oscillations controlled by the *original* weights. In the route giving the stated numerical bound, Section C.4 gives mean at most $7/8$ and port oscillation at most $2h_{ig}$. A bounded-differences exponential-moment estimate then bounds each coefficient-free branch by $e^{-r/(128C*sw)}$. Sparse correction sets are summed using this estimate and their binomial count. Dense sets use the additional joint penalty

$$\prod_i \|K_i\|_1 \leq \exp\left[-r\Phi\left(\frac{sw}{r}\right)\right] \quad (r \geq 1, s > 0),$$

obtained from joint insertion and Jensen's inequality. Here $sw \leq C * sr$ by the total source budget. For a suitable absolute choice of w , the penalty offsets the count of dense sets and the factor 8^s . The empty correction set and $r = 0$ case are treated separately in the detailed argument. The independent resource is bounded jointly in trace norm at every step; it is never normalized or factored over its ports.

This proves a bound $K_0 e^{-ar}$ in the number of one-input sources. To return to the original n inputs, fix a sufficiently large constant integer cutoff $L \geq 2$ depending only on a . Expand each first-layer component with $k_g > L$ inputs as its deleted baseline plus the rank-at-most-two correction of trace norm at most 42^{-k_g} . The baseline contributes k_g singleton sources; the correction contributes one small resource operator after all its input variables have already been signed and averaged. In every other input-containing component, keep one variable and condition the others. If these

components contain m inputs and give r_0 sources, then $m \leq Lr_0$. With $b = \min\{a, (\log 2)/2\}$, summing all correction subsets gives

$$K_0 e^{-br_0} \prod_{g: k_g > L} \left(e^{-bk_g} + 4 \cdot 2^{-k_g} \right).$$

The choice of L in Theorem B.4 ensures that each factor is at most $e^{-bk_g/2}$; hence the expression is bounded by $K_0 e^{-(b/L)n}$. Product effects on *physical qubits* are essential here: they still factor over ports when a baseline component splits into singleton sources. Finally, pulling back the one-qubit output expresses the third-layer observable as a linear combination of precisely such positive product effects, with total absolute coefficient mass at most six.

1.6 Organization of the remaining sections

Section 2 specifies the model, normalizations, formal theorems, and bounded-error consequences. Section 3 proves the robust first-layer restriction. Section 4 establishes the source geometry, mixedness budget, retention majorant, and insertion inequality. Section 5 proves the basic source-network concentration bound. Section 6 handles the measured output and completes the shorter correlation estimate, while Section 7 gives the explicit circuits with exponentially small nonzero correlation.

Appendix A proves the standard concentration inequality in the form used here. Appendix B develops retention after joint insertion, the sparse/dense heavy-gate sum, and the exact first-layer correction, thereby proving the main exponential theorem. Appendix C gives alternative retention arguments, including transfer of the original monotone majorant. Appendix D replaces the small-trigger cutoff by a convex joint-loss estimate and improves the explicit constants.

2 Scope, normalization, and statements

Circuit model. Fix integers $n \geq 1$ and $m \geq 0$. The circuit acts on $N = n + m$ qubits, with initial state

$$\rho_x^{\text{in}} = |x\rangle\langle x| \otimes \sigma_{\text{anc}}, \quad x \in \{0, 1\}^n, \quad \sigma_{\text{anc}} = \bigotimes_{j=1}^m \sigma_j,$$

where each σ_j is a one-qubit density operator independent of x . For $m = 0$ the ancillary factor is the scalar one. Thus there is exactly one supplied computational-basis copy of each input bit. The number m is any finite integer; no bound in terms of n is imposed on it.

Every multiqubit gate is a generalized Toffoli gate: on computational-basis states it flips its target precisely when all of its control bits are one. No other multiqubit gates are allowed. More explicitly, the circuit unitary has the form

$$U_C = V_3 \mathcal{L}_3 V_2 \mathcal{L}_2 V_1 \mathcal{L}_1 V_0,$$

where each V_j is a tensor product of one-qubit unitaries, and each $\mathcal{L}_j$ is a layer of multiqubit generalized Toffoli gates with pairwise disjoint supports on *all* wires, including controls. A layer may be empty and then acts as the identity. Depth counts only the nonempty multiqubit layers, so this notation includes circuits of depth zero, one, two, or three. Gates may connect arbitrary wires; no geometric locality is imposed.

A fixed output wire q , which may be an input or an ancillary wire, is measured once at the end with a binary positive operator-valued measure (M_0, M_1) :

$$M_0, M_1 \geq 0, \quad M_0 + M_1 = I_2.$$

The output bit is the measurement label. All other wires are ignored and may contain arbitrary garbage; the circuit need not preserve the input or restore the ancillas. There are no intermediate measurements, postselection, input-dependent gate choices, or additional classical processing of other output wires. The unitary, initial ancillary state, output wire, and measurement are fixed independently of x . They may depend on n . All Hilbert spaces are finite-dimensional, and every circuit is finite, but there is no polynomial-size or uniformity requirement.

Reflection normalization. On its support S , a generalized Toffoli gate is locally equivalent to a rank-one product reflection

$$D = I_S - 2\Pi, \quad \Pi = \bigotimes_{q \in S} |v_q\rangle\langle v_q|, \quad (1)$$

where every v_q is a unit vector. Operators on S are extended by the identity on the complementary wires when used in the full circuit. In particular, Π has rank one on the gate's support, not necessarily on the full register. Since $\Pi^2 = \Pi = \Pi^*$, the gate satisfies $D^* = D$ and $D^2 = I_S$.

Indeed, for control set R and target t , the usual controlled- X gate is

$$I_{R \cup \{t\}} - 2|1 \cdots 1\rangle\langle 1 \cdots 1|_R \otimes |-\rangle\langle -|_t, \quad |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}.$$

Conversely, a tensor product of one-qubit unitaries maps the displayed trigger vector to any product vector $\bigotimes_{q \in S} v_q$, so conjugation gives (1). The local changes of basis for disjoint gates combine into one-qubit layers and can be absorbed into the neighboring V_j . Thus the reflection description preserves the allowed supports and the number of multiqubit layers. A one-qubit reflection can be included in a V_j and need not be counted as a multiqubit gate. We retain arbitrary local unitaries at layer boundaries; the standard normal form also permits absorbing them further; see [10, Proposition 2.1] and [17].

Correlation and measurement convention. Let $\rho_{C,q}(x)$ be the reduced density operator on the output wire after applying U_C . Write $p_C(b | x) = \text{Tr}(M_b \rho_{C,q}(x))$ for $b \in \{0, 1\}$. The sign observable of the measurement is

$$A = M_0 - M_1 = 2M_0 - I_2, \quad A = A^*, \quad \|A\| \leq 1,$$

where $\|\cdot\|$ denotes operator norm. No tracelessness or unitarity of A is assumed. Conversely, every Hermitian one-qubit contraction defines such a measurement through $M_0 = (I_2 + A)/2$ and $M_1 = (I_2 - A)/2$. Hence the model includes all binary measurements on the designated qubit, not only projective measurements in a chosen basis.

Set $a_C(x) = p_C(0 | x) - p_C(1 | x) = \text{Tr}(A \rho_{C,q}(x))$ and

$$\beta(C) = \mathbb{E}_{x \text{ uniform on } \{0,1\}^n} \chi_n(x) a_C(x), \quad \chi_n(x) = (-1)^{x_1 + \cdots + x_n}. \quad (2)$$

Both $|a_C(x)| \leq 1$ and $|\beta(C)| \leq 1$. If X is uniform and Y is the measured output of C on X , then

$$\mathbb{P}[Y = \text{PARITY}_n(X)] = \frac{1 + \beta(C)}{2}.$$

For each fixed input x , the corresponding conditional success is $(1 + \chi_n(x) a_C(x))/2$, which proves the identity by averaging. Thus average error at most ε is equivalent to $\beta(C) \geq 1 - 2\varepsilon$. Error at most ε on *every* input implies the same inequality, but the converse need not hold. Relabelling the two outcomes changes the sign of $\beta(C)$, so the best uniform-average advantage after such a relabelling is $|\beta(C)|/2$.

Reduction to pure product ancillas. It suffices to prove the bounds for pure product initialization. Spectrally decomposing each σ_j gives a finite convex decomposition

$$\sigma_{\text{anc}} = \sum_z w_z |\alpha_z\rangle\langle\alpha_z|, \quad w_z \geq 0, \quad \sum_z w_z = 1,$$

in which every α_z is a product vector and the weights are independent of x . For fixed U_C and measurement, correlation is linear in the initial density operator. If C_z denotes the same circuit and measurement with initial ancillas $|\alpha_z\rangle$, then

$$\beta(C) = \sum_z w_z \beta(C_z), \quad |\beta(C)| \leq \sum_z w_z |\beta(C_z)|.$$

Every bound below is uniform in these product vectors, so the same bound holds for the mixed product state. No purification or additional circuit depth is needed. We make this reduction before using the pure-source identities in Section 4.

Theorem 2.1 (Exponential parity-correlation bound). *There are absolute constants $K_0 \geq 1$ and $\kappa > 0$ such that, for every integer $n \geq 1$, every finite number of ancillas, and every circuit C in the stated model on those qubits,*

$$|\beta(C)| \leq 6K_0 e^{-\kappa n}.$$

The constants do not depend on gate count, ancillary width, gate arities, local bases, the initial product state, or the chosen one-qubit binary measurement. In particular, one has the explicit specialization

$$|\beta(C)| \leq \min\{1, 1800 \exp(-2^{-27}n)\}.$$

Proof. The full exponential argument is developed in Appendices B–D. Theorem B.4 bounds the full parity coefficient of every positive physical-qubit product effect after two layers, without restricting first-layer input arities. The local pullback in the proof of Lemma 6.1, which does not require first-layer cleanup, expresses the third-layer sign observable as a linear combination of such effects with total absolute coefficient mass at most six. This yields Corollary B.5. In particular, the argument applies to every Hermitian one-qubit contraction A defined above.

For the numerical specialization, the original-majorant transfer in Section C.4 and the convex joint-loss argument in Appendix D give a source rate $a = 2^{-21}$ and prefactor $K_0 < 300$. The first-layer cutoff in Theorem B.4 is then $L = 48$, giving $\kappa = a/L > 2^{-27}$. Corollary D.2 consequently gives $|\beta(C)| \leq 1800e^{-2^{-27}n}$. Combine this with $|\beta(C)| \leq 1$ and the mixed-product reduction above. $\square$

The argument using Appendix B alone, without the alternative retention and convex refinements, gives the weaker but valid specialization $|\beta(C)| \leq 30000 \exp(-2^{-38}n)$; see Section B.4. The stronger numerical bound above uses the additional dependencies just stated. Neither explicit constant is intended to give a sharp finite-input threshold.

We also retain a shorter quantitative bound whose proof uses only the main text and Appendix A. It already establishes size-independent bounded-error impossibility. Set $C * s = \frac{1}{2} + \frac{2}{e}$ and, for real $r \geq 1$, define

$$\Delta(r) = 8C * s \sqrt{r} e^{-\sqrt{r}} + (7/8)^{r/2} + e^{-\sqrt{r}/(512C*s)}. \quad (3)$$

Allowing real r makes it possible to substitute the lower bound $n/(k-1)$ for an integer number of surviving inputs. All logarithms without a subscript are natural.

Theorem 2.2 (Short quantitative bound). *For every integer $n \geq 1$, every circuit C in the stated model on n inputs, and every integer k with $2 \leq k \leq n + 1$,*

$$|\beta(C)| \leq 4\sqrt{\frac{n}{k}} 2^{-k} + 6\Delta\left(\frac{n}{k-1}\right). \quad (4)$$

Consequently, there are absolute constants $K, c > 0$ such that

$$|\beta(C)| \leq K e^{-cn^{1/3}}. \quad (5)$$

These estimates are uniform in the same circuit parameters as Theorem 2.1.

Proof. The restriction and concentration argument is proved in Sections 3–6; we summarize the deductions needed for this statement. Lemma 3.1 bounds the first-layer deletion cost by $4\sqrt{(n/k)2^{-k}}$. The survivor count (6) gives an integer $r \geq \lceil n/(k-1) \rceil$, and Lemma 6.1 bounds every restricted coefficient by $6\Delta(r)$. Averaging with the parity signs of the fixed inputs restores the original coefficient of the deleted circuit. The function Δ is decreasing on $[1, \infty)$: writing $y = \sqrt{r} \geq 1$, its first term is a positive constant times ye^{-y} , and its other terms decrease directly. The triangle inequality therefore proves (4).

For $n \geq 8$, take $k = \lceil n^{1/3} \rceil$. This choice is admissible and satisfies $n/(k-1) \geq n^{2/3}$. Equation (17) in the detailed proof then bounds every term by an absolute constant times $e^{-cn^{1/3}}$, for example with $c = 1/(1024C * s)$, after absorbing the polynomial prefactors. Enlarging K covers $1 \leq n < 8$ using $|\beta(C)| \leq 1$. $\square$

Corollary 2.3 (Fixed bounded error). *For every fixed $0 \leq \varepsilon < 1/2$ there is an integer $n_0(\varepsilon)$ such that no circuit in the stated model computes parity on $n \geq n_0(\varepsilon)$ inputs with uniform-input average error at most ε , whatever its finite ancillary width or gate count. In particular, no such circuit has error at most ε on every input. One may take*

$$n_0(\varepsilon) = 1 + \left\lceil 2^{27} \log \frac{1800}{1-2\varepsilon} \right\rceil.$$

Proof. Put $\theta = 2^{27} \log(1800/(1-2\varepsilon))$. For $n \geq n_0(\varepsilon) = 1 + \lfloor \theta \rfloor$ one has $n > \theta$, including when θ is an integer. Thus Theorem 2.1 gives

$$\beta(C) \leq |\beta(C)| \leq 1800e^{-n/2^{27}} < 1800e^{-\theta/2^{27}} = 1 - 2\varepsilon.$$

This contradicts the necessary correlation for average error at most ε . A worst-case guarantee would imply that average guarantee. Alternatively, existence of some threshold follows already from (5); the displayed threshold uses the stronger exponential estimate. $\square$

The conclusion is a size-independent impossibility for sufficiently large n , not merely a lower bound on the size of a successful circuit. It is compatible with exact small-input circuits, such as those in Proposition 7.1. Unrestricted ancillary *width* here does not grant initially entangled advice, free fanout, multiple supplied input copies, postselection, or joint classical processing of many measured outputs.

3 Robust removal of large-input first-layer gates

This section gives the restriction step used in the short bound, Theorem 2.2. We first control the average error from deleting first-layer gates that touch many input wires, then retain one variable input per remaining component. This delete-and-restrict strategy adapts [17, Section 5.1, Claims 5.2–5.3]; the estimate below handles all selected disjoint gates together and concerns the signed parity coefficient, not a Fourier-tail weight.

Work in the product-reflection representation of Section 2. Here “first layer” means the indexed stage $\mathcal{L}_1$, which may be empty, rather than the first nonempty layer. Deleting a gate means replacing its reflection by the identity on the same wires, with all boundary one-qubit unitaries, initial states, and the final measurement unchanged.

Lemma 3.1 (Mean deletion error). *Let C be a circuit in the stated model, let H be any set of its first-layer multiqubit gates, and let $r_g \geq 0$ be the number of original input wires in $g \in H$. If C^- is obtained by deleting the gates in H , then*

$$|\beta(C) - \beta(C^-)| \leq 4 \sqrt{\sum_{g \in H} 2^{-r_g}}.$$

The bound includes $H = \emptyset$, with the empty sum equal to zero.

Proof. First take pure product ancillas. Immediately before $\mathcal{L}_1$, the state $|w_x\rangle = V_0(|x\rangle \otimes |\alpha\rangle)$ is a unit product vector over individual wires. For $g \in H$, write S_g for its support, $|w_{x,g}\rangle$ for the corresponding factor, and

$$D_g = I - 2\Pi_g, \quad p_g(x) = \langle w_{x,g} | \Pi_g | w_{x,g} \rangle \in [0, 1], \quad U_H = \prod_{g \in H} D_g.$$

The supports S_g are pairwise disjoint, including control wires. Thus U_H is a Hermitian unitary and

$$\langle w_x | U_H | w_x \rangle = \prod_{g \in H} (1 - 2p_g(x)).$$

For any $a_1, \dots, a_h \in [-1, 1]$, the telescoping identity gives

$$1 - \prod_{j=1}^h a_j = \sum_{j=1}^h (1 - a_j) \prod_{\ell < j} a_\ell \leq \sum_{j=1}^h (1 - a_j).$$

Indeed, each prefix product is at most one and each $1 - a_j$ is nonnegative; the prefix need not be nonnegative. Enumerating $H = \{g_1, \dots, g_h\}$ and using $a_j = 1 - 2p_{g_j}(x)$, we obtain

$$\|(U_H - I)|w_x\rangle\|^2 = 2 \left(1 - \prod_{g \in H} (1 - 2p_g(x)) \right) \leq 4 \sum_{g \in H} p_g(x).$$

Empty products equal one, so these identities also cover $H = \emptyset$.

Let $|\psi_x\rangle$ and $|\psi_x^-\rangle$ be the full final premeasurement pure states of C and C^- . The remaining first-layer gates commute with U_H and are common to both circuits, as are all subsequent unitaries. Consequently

$$d_x := \|\psi_x - \psi_x^-\| = \|(U_H - I)|w_x\rangle\|.$$

Let $V_{0,q}$ be the initial one-qubit unitary on input wire q . Averaging its uniformly random input bit gives

$$V_{0,q}(I_2/2)V_{0,q}^* = I_2/2.$$

Uniform input bits are independent. If $\Pi_{g,\text{anc}}$ is the product-projector factor on the ancillary wires of g and τ_g is their state immediately before the layer, then

$$\mathbb{E}_x p_g(x) = 2^{-r_g} \text{Tr}(\Pi_{g,\text{anc}} \tau_g) \leq 2^{-r_g}.$$

Here the ancillary trace is one when there are no ancillary wires. The inequality also holds when $r_g = 0$, though it need not be small.

For $\rho_x = |\psi_x\rangle\langle\psi_x|$ and $\rho_x^- = |\psi_x^-\rangle\langle\psi_x^-|$, the pure-state trace distance satisfies

$$T(\rho_x, \rho_x^-) := \frac{1}{2} \|\rho_x - \rho_x^-\|_1 = \sqrt{1 - |\langle\psi_x|\psi_x^-\rangle|^2} \leq d_x.$$

For the last inequality, putting $c = \langle\psi_x|\psi_x^-\rangle$ gives $d_x^2 - (1 - |c|^2) = |1 - c|^2 \geq 0$. The full-register sign observable is the measured one-qubit contraction tensored with identities, so trace-norm duality gives $|a_C(x) - a_{C^-}(x)| \leq 2T(\rho_x, \rho_x^-) \leq 2d_x$. The triangle inequality and Cauchy-Schwarz now yield

$$\begin{aligned} |\beta(C) - \beta(C^-)| &\leq \mathbb{E}_x |a_C(x) - a_{C^-}(x)| \leq 2\mathbb{E}_x d_x \\ &\leq 2\sqrt{\mathbb{E}_x d_x^2} \leq 4\sqrt{\sum_{g \in H} 2^{-r_g}}. \end{aligned}$$

Finally, apply the pure-product decomposition from Section 2 to the common initial ancillary state of C and C^- . Their correlation difference is linear in that state, and the displayed bound is uniform in each pure product component. Convexity therefore proves the same bound for product mixed ancillas. $\square$

Restriction and survivor count. Fix an integer $k \geq 2$ and let H consist of all first-layer gates containing at least k original inputs. Disjointness gives $k|H| \leq \sum_{g \in H} r_g \leq n$, hence

$$|\beta(C) - \beta(C^-)| \leq 4\sqrt{|H|2^{-k}} \leq 4\sqrt{\frac{n}{k}2^{-k}}.$$

A *first-layer component* of C^- is the support of a remaining first-layer gate, or a singleton wire not in any such gate. These components partition all physical wires. In particular, every wire of a deleted gate is now a singleton at this stage; later gates on that wire are not deleted.

Choose one original input from each remaining gate containing inputs, and also choose every original input in a singleton component. This defines a set $J \subseteq [n] = \{1, \dots, n\}$ independently of the input assignment. Each input-containing component contains between one and $k-1$ original inputs and contributes exactly one member to J . Therefore $n \leq (k-1)|J|$, and

$$r = |J| \geq \left\lceil \frac{n}{k-1} \right\rceil. \quad (6)$$

In particular $1 \leq r \leq n$. When $k > n$, no gate is deleted; the same counting argument still applies.

For each $z \in \{0, 1\}^{[n] \setminus J}$, let C_z^- be C^- with those original inputs fixed to z , and with the bits $y \in \{0, 1\}^J$ left variable. No wire or further gate is removed. The initial state is, up to ordering of wires,

$$|y\rangle\langle y| \otimes |z\rangle\langle z| \otimes \sigma_{\text{anc}}.$$

Thus the fixed input wires are additional product ancillas independent of y ; their original one-qubit gates remain in the circuit. The output wire and measurement also remain unchanged, even if that wire was among the fixed inputs. Each C_z^- is therefore a circuit in the same model with r variable inputs and $m + n - r$ ancillas, and each first-layer component contains at most one variable input.

After the pure-product reduction of Section 2, the r input-containing components supply the independent one-input sources of Section 4. Components with no variable input produce an input-independent state, possibly entangled within a component. Grouping all of them gives a resource density operator tensor-independent of the sources, as allowed in Theorem 5.1. One-qubit unitaries at the next boundary preserve this factorization. The resource is the scalar one if there are no such wires.

Exact coefficient reconstruction. For $S \subseteq [n]$, write $\chi_S(u) = (-1)^{\sum_{i \in S} u_i}$, with $\chi_\emptyset = 1$. For the restricted circuit, its parity correlation is

$$\beta(C_z^-) = \mathbb{E}_{y \in \{0,1\}^J} \chi_J(y) a_{C^-}(y, z),$$

where (y, z) denotes the full input with those coordinates; all expectations in this paragraph are uniform. Factoring $\chi_{[n]}(y, z) = \chi_J(y) \chi_{[n] \setminus J}(z)$ gives the exact identity

$$\beta(C^-) = \mathbb{E}_z [\chi_{[n] \setminus J}(z) \beta(C_z^-)]. \quad (7)$$

Consequently, any common bound $|\beta(C_z^-)| \leq B$ valid for *every* assignment z implies $|\beta(C^-)| \leq B$. The identity includes $J = [n]$: the complement then has a single empty assignment and parity sign one. No random choice of representatives, favourable restriction, output relabeling, or preservation of worst-case correctness is required. Section 6 applies this identity with $B = 6\Delta(r)$.

4 Geometry of one-input sources

Pure sources and ports. We work after the pure-product reduction in Section 2. In particular, the spectral statements below apply to each pure product component, not directly to an average over mixed ancillary initializations. A source consists of one variable input wire d and any finite number of ancillary wires in a pure product state, acted on by one product reflection. Untouched input singletons are also included, by the representation described below.

Partition the source's wires into nonempty ports indexed by $\mathcal{J} = \{0, 1, \dots, s\}$, where $s \geq 0$ is an integer and the *owner* port 0 contains d . All other ports contain only ancillary wires. Ports may have arbitrary widths, and the owner may itself contain ancillary wires. For $S \subseteq \mathcal{J}$, $\text{Tr}_{\mathcal{J} \setminus S}$ traces all wires in the ports outside S ; the empty retained register is one-dimensional. Complements such as A^c below are taken in $\mathcal{J}$.

Choose the data basis so that the reflection's trigger on d is $|1\rangle$. Then

$$G = I - 2|1\rangle\langle 1|_d \otimes \Pi_{\text{anc}}, \quad \Pi_{\text{anc}} = \bigotimes_{q \text{ ancillary}} |v_q\rangle\langle v_q|,$$

where each v_q is a unit vector. The two data input vectors u_0, u_1 are orthonormal, and the ancillary input α is a unit product vector. Define, for $x \in \{0, 1\}$,

$$\psi_x = G(u_x \otimes \alpha), \quad F = \frac{|\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|}{2}, \quad \bar{\rho} = \frac{|\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1|}{2}.$$

Unitarity of G makes ψ_0, ψ_1 orthonormal. Hence F is Hermitian, $\text{Tr } F = 0$, and $\|F\|_1 = 1$, while $\bar{\rho}$ is a density operator. If there are no ancillary wires, the ancillary vector and projector are the

scalar one. An untouched singleton is represented by the formal reflection $G = Z_d$ and input vectors $u_x = Z_d w_x$, where $Z_d = |0\rangle\langle 0| - |1\rangle\langle 1|$ and w_0, w_1 are its actual orthonormal output vectors. This adds no physical gate.

We prove the formulas in these source coordinates. Initial one-qubit unitaries are already included in u_0, u_1 and α . One-qubit unitaries after the source reflection act within ports. They conjugate each retained operator by a unitary on its retained ports, preserving the marginal eigenvalues and retained trace norms. For an inserted operator on a traced port, the same change of coordinates conjugates that operator, preserving its rank and norm. Thus the conclusions also hold with those boundary unitaries present.

For port j , write $\bar{\rho}_j$ for its marginal and set

$$\lambda_j = \lambda_{\max}(\bar{\rho}_j), \quad h_j = 1 - \lambda_j.$$

Recall that $C * s = \frac{1}{2} + \frac{2}{e}$.

Lemma 4.1 (Source budget and retention). *The pure source above has the following properties.*

- (i) $h_0 = 1/2$, $0 \leq h_j \leq 1/2$ for every port, and $\sum_{j \in \mathcal{J}} h_j \leq C * s$.
- (ii) There is $f : 2^{\mathcal{J}} \rightarrow [0, 1]$, nondecreasing under inclusion, such that $\|\text{Tr}_{\mathcal{J} \setminus S} F\|_1^2 \leq f(S)$ for every $S \subseteq \mathcal{J}$. If each port is retained independently with probability $1/2$, then $\mathbb{E}f \leq 3/4$.
- (iii) For every $S \subseteq \mathcal{J}$ and $j \in \mathcal{J} \setminus S$, $|f(S \cup \{j\}) - f(S)| \leq 4h_j$.

Proof. Marginal spectra and the weight budget. For each ancillary-only port j , let α_j be its initial unit vector and let Π_j be the tensor product of the trigger projectors on its wires. Put

$$p_j = \langle \alpha_j | \Pi_j | \alpha_j \rangle, \quad R_j = I_j - 2\Pi_j.$$

Here R_j is one reflection on the whole port, not a tensor product of one-qubit reflections. Let t_0 be the product of the trigger probabilities of the ancillary wires in the owner, and set

$$Q = \prod_{j=1}^s p_j, \quad P = t_0 Q = \langle \alpha | \Pi_{\text{anc}} | \alpha \rangle.$$

All these probabilities lie in $[0, 1]$; empty products equal one.

The uniform mixture of u_0, u_1 is $I_d/2$, so

$$\bar{\rho} = \frac{1}{2}|0\rangle\langle 0| \otimes |\alpha\rangle\langle \alpha| + \frac{1}{2}|1\rangle\langle 1| \otimes R_{\text{anc}}|\alpha\rangle\langle \alpha|R_{\text{anc}}, \quad R_{\text{anc}} = I - 2\Pi_{\text{anc}}. \quad (8)$$

In the owner marginal, the data-zero block is a pure state with weight $1/2$: tracing other ports preserves purity of the owner's initial ancillary factor. The orthogonal data-one block is positive and has trace $1/2$, so none of its eigenvalues exceeds $1/2$. Thus $\lambda_0 = 1/2$ and $h_0 = 1/2$. This argument does not require the owner marginal to have rank at most two.

If $P = 0$, then $\Pi_{\text{anc}}\alpha = 0$ and G acts as the identity on the whole logical input subspace. All ancillary-only marginals are pure. Tracing out the owner makes F zero, while retaining it leaves trace norm one, since the remaining ancillary factors are pure and the data states are orthogonal. Consequently $f(S) = \mathbf{1}_{\{0 \in S\}}$ is the exact squared retained norm. Its mean is $1/2$, and its only nonzero toggle has size $1 \leq 4h_0$. All three claims follow in this case.

Assume henceforth $P > 0$, so t_0 and every p_j are positive. For an ancillary-only port j , let $r_j = P/p_j \in (0, 1]$. Expanding (8) and tracing the other wires gives

$$\bar{\rho}_j = (1 - r_j/2)|\alpha_j\rangle\langle \alpha_j| + (r_j/2)R_j|\alpha_j\rangle\langle \alpha_j|R_j.$$

This is a mixture of two pure states, so it has rank at most two and $\lambda_j \geq 1/2$. Its two potentially nonzero eigenvalues are $1-h_j$ and h_j ; the latter is zero in the rank-one case. Since $\langle \alpha_j | R_j | \alpha_j \rangle = 1-2p_j$, their product is

$$h_j(1-h_j) = \frac{r_j}{2} \left(1 - \frac{r_j}{2}\right) (1 - (1-2p_j)^2) = P(1-p_j)(2-r_j).$$

The lower bound below follows from $2-r_j \geq 1$ and $h_j(1-h_j) \leq h_j$:

$$P(1-p_j) \leq h_j \leq 2P(1-p_j). \quad (9)$$

For the upper bound, the Rayleigh quotient at α_j is $1-2P(1-p_j)$, which is at most λ_j . Using $1-p \leq -\log p$ for $0 < p \leq 1$, $P \leq Q$, and $-P \log P \leq 1/e$, we obtain

$$\sum_{j=1}^s h_j \leq 2P \sum_{j=1}^s (1-p_j) \leq -2P \log Q \leq -2P \log P \leq 2/e.$$

Together with $h_0 = 1/2$ and the ancillary rank bound, this proves (i).

Exact retention when the owner is erased. Put $\delta = |\langle 1|u_0\rangle|^2 - |\langle 1|u_1\rangle|^2$, so δ is real and $|\delta| \leq 1$. Tracing the data wire eliminates its off-diagonal matrix entries:

$$\text{Tr}_d F = \frac{\delta}{2} (R_{\text{anc}} |\alpha\rangle \langle \alpha| R_{\text{anc}} - |\alpha\rangle \langle \alpha|).$$

For a retained set $A \subseteq \{1, \dots, s\}$ of ancillary ports, define

$$\begin{aligned} \alpha_A &= \bigotimes_{j \in A} \alpha_j, & \rho_A &= |\alpha_A\rangle \langle \alpha_A|, & \Pi_A &= \bigotimes_{j \in A} \Pi_j, & R_A &= I_A - 2\Pi_A, \\ P_A &= \prod_{j \in A} p_j, & P_I &= t_0 \prod_{j \in \{1, \dots, s\} \setminus A} p_j, & P &= P_A P_I. \end{aligned}$$

In particular, R_A is the joint reflection on all retained ancillary ports. Expanding R_{anc} and tracing the erased ancillary factors gives the exact signed operator

$$\text{Tr}_{A^c} F = \frac{\delta P_I}{2} (R_A \rho_A R_A - \rho_A).$$

For unit vectors u, v , their projector difference has trace norm $2\sqrt{1 - |\langle u|v\rangle|^2}$. Its nonzero eigenvalues, when present, are the opposite pair $\pm\sqrt{1 - |\langle u|v\rangle|^2}$; if the vectors agree up to phase, the difference is zero. Since $\langle \alpha_A | R_A | \alpha_A \rangle = 1 - 2P_A$, we obtain

$$D(A)^2 := \|\text{Tr}_{A^c} F\|_1^2 = 4\delta^2 P_I^2 P_A (1 - P_A) = 4\delta^2 P^2 (P_A^{-1} - 1). \quad (10)$$

The inverse is used only in the current case $P > 0$. For $A = \emptyset$, one has $P_A = 1$, the scalar reflection $R_A = -1$, and zero retained signed operator, as required.

The majorant, its mean, and its oscillations. Define

$$f(S) = \begin{cases} 1, & 0 \in S, \\ D(S)^2, & 0 \notin S. \end{cases}$$

Every partial trace of F is a half-difference of density operators, so its trace norm is at most one. Hence f is a majorant taking values in $[0, 1]$. It is not asserted to equal the squared retained

norm when the owner is retained. For owner-free sets $A \subseteq B$, the operator retained on A is a partial trace of the operator retained on B . Trace-norm contractivity on Hermitian operators proves $D(A) \leq D(B)$. Adding the owner sets f to one, so f is nondecreasing under inclusion.

Conditioned on erasing the owner, the remaining port coins are still independent and fair. Taking the expectation of P_A^{-1} in (10) and factoring over these coins gives

$$\mathbb{E}_A D(A)^2 = 4\delta^2 t_0^2 Q \left(\prod_{j=1}^s \frac{1+p_j}{2} - Q \right).$$

For $a, b \in [0, 1]$, the inequality $(1+a)(1+b)/4 \leq (1+ab)/2$ is equivalent to $(1-a)(1-b) \geq 0$. Induction, including the empty-product case, yields

$$\prod_{j=1}^s \frac{1+p_j}{2} \leq \frac{1 + \prod_{j=1}^s p_j}{2} = \frac{1+Q}{2}.$$

Thus $\mathbb{E}_A D(A)^2 \leq 2\delta^2 t_0^2 Q(1-Q) \leq 1/2$. Averaging the owner's independent fair coin gives $\mathbb{E}f = \frac{1}{2} + \frac{1}{2}\mathbb{E}_A D(A)^2 \leq 3/4$, proving (ii).

If the owner is erased and $j \notin A$ is an ancillary port, (10) gives

$$f(A \cup \{j\}) - f(A) = 4\delta^2 \frac{P^2}{P_{Ap_j}} (1-p_j) \leq 4P(1-p_j) \leq 4h_j.$$

Here $P/(P_{Ap_j}) \leq 1$ is the product of the still-erased ancillary trigger probabilities. If the owner is retained, an ancillary toggle changes nothing. Adding or removing the owner changes f by at most $1 \leq 4h_0$, and removing an ancillary port reverses the corresponding addition. This proves (iii) in every case. $\square$

Lemma 4.2 (Single-port insertion). *For the pure source above and any rank-at-most-one operator B on port j ,*

$$\|\text{Tr}_j[(B \otimes I)F]\|_1 \leq \|B\| \lambda_j.$$

No Hermiticity or positivity of B is required.

Proof. Let $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ on a two-dimensional logical input space, and let W be the isometry $W|x\rangle = \psi_x$. Then $F = WZW^*/2$ and $\bar{\rho} = WW^*/2$. For a unit vector u on port j , put $W_u = (\langle u| \otimes I)W$. Writing $\|X\| * 2 = (\text{Tr } X^* X)^{1/2}$ for the Hilbert–Schmidt norm gives

$$\|W_u\| * 2^2 = 2\langle u|\bar{\rho}_j|u\rangle \leq 2\lambda_j.$$

If $B \neq 0$, its singular-value decomposition has the form $B = b|u\rangle\langle v|$, with u, v unit vectors and $b = \|B\| > 0$. The partial trace, in this order, is

$$\text{Tr}_j[(B \otimes I)F] = \frac{b}{2} W_v Z W_u^*.$$

Schatten Hölder and unitarity of Z imply

$$\|\text{Tr}_j[(B \otimes I)F]\|_1 \leq \frac{b}{2} \|W_v Z\| * 2 \|W_u^*\| * 2 = \frac{b}{2} \|W_v\| * 2 \|W_u\| * 2 \leq b\lambda_j.$$

The zero operator is immediate. The same proof includes a one-dimensional complementary register when the port is the whole source. $\square$

Remark 4.3 (A necessary caution). Individual insertion penalties cannot, in general, be multiplied for different ports of the same source. Apply a CNOT to $|x\rangle \otimes |0\rangle$ and take its two wires as singleton ports. Then

$$F = (|00\rangle\langle 00| - |11\rangle\langle 11|)/2, \quad \bar{\rho} = (|00\rangle\langle 00| + |11\rangle\langle 11|)/2.$$

Both single-port marginals of $\bar{\rho}$ are $I_2/2$, so $\lambda_0 = \lambda_1 = 1/2$, whereas

$$|\text{Tr}(|00\rangle\langle 00|F)| = 1/2 > \lambda_0\lambda_1 = 1/4.$$

There is no contradiction with a single insertion on the joint port: the joint marginal has largest eigenvalue $1/2$. The rank restriction is also essential. For the rank-two choice $B = I_2$ on port 0, one has $\|\text{Tr}_0 F\|_1 = 1 > \|I_2\| \lambda_0 = 1/2$.

The basic network proof estimates each gate deletion separately on the original sources. For joint insertions, Lemma B.1 in Appendix B uses the joint marginal rather than a product of individual marginal penalties.

5 Two-layer product-event concentration

We now combine the source estimates with the second layer. The source states are tensor-independent, but a second-layer gate can couple several sources. The proof first deletes gates with large total source weight, and then applies concentration to an exact random expansion of the remaining positive effect. Only the source estimates in Section 4 and the bounded-differences inequality in Appendix A are needed.

Theorem 5.1 (Basic source-network estimate). *Let $r \geq 1$ be an integer. For each $i \in \{1, \dots, r\}$, take a one-input source as in Section 4, with conditional state $\rho_{i,x_i} = |\psi_{i,x_i}\rangle\langle\psi_{i,x_i}|$. Let σ be an arbitrary input-independent resource density operator, tensor-independent of all input sources. It may be internally entangled; an absent resource is the scalar one. The joint state is therefore $\sigma \otimes \bigotimes_{i=1}^r \rho_{i,x_i}$.*

Apply a second layer U_2 of product reflections with pairwise disjoint supports. For each source, its ports are its nonempty intersections with these gate supports, together with one port of all its idle wires if that set is nonempty. Treat the entire resource as one additional group and define its ports in the same way, without assuming that its state factors over them. Let E be a tensor product of positive contractions over all these ports; a port factor need not factor further over its individual qubits. Define

$$q(x) = \text{Tr} \left[EU_2 \left(\sigma \otimes \bigotimes_{i=1}^r \rho_{i,x_i} \right) U_2^* \right], \quad \alpha = \mathbb{E}_x(-1)^{|x|} q(x).$$

Here x is uniform on $\{0, 1\}^r$, and $0 \leq q(x) \leq 1$. For every real $w \geq 1$, the full parity coefficient satisfies

$$|\alpha| \leq 8 \frac{C * sr}{w} e^{-w} + (7/8)^{r/2} + e^{-r/(512C*sw)}. \quad (11)$$

In particular $|\alpha| \leq \Delta(r)$.

Proof. Network representation. One-qubit unitaries before and after the second layer act within ports. They can be absorbed into the source states (including σ) and event factors, respectively, using the covariance established in Section 4. Put

$$F_i = \frac{\rho_{i,0} - \rho_{i,1}}{2}, \quad \Sigma = \mathbb{E}_x(-1)^{|x|} \left(\sigma \otimes \bigotimes_{i=1}^r \rho_{i,x_i} \right) = \sigma \otimes \bigotimes_{i=1}^r F_i.$$

Thus $\alpha = \text{Tr}[(U_2^* E U_2) \Sigma]$, with no additional normalization factor.

Represent each second-layer gate by a *gate vertex*. Give each nonempty idle port its own *private vertex*, including a resource idle port when present. These vertex wire sets partition the whole register. At a gate vertex g , write

$$E_g = D_g P_g D_g, \quad D_g = I - 2\Pi_g,$$

where P_g is the tensor product of the original event factors on its ports. It is a positive contraction, not necessarily a projector. The trigger Π_g also factors over these ports, with a rank-one projector on every present port. At a private vertex, E_g is just its port's event factor. Consequently $U_2^* E U_2 = \bigotimes_g E_g$, and every E_g is a positive contraction. Each source has at most one port at a given vertex.

Heavy-gate deletion. For a port of source i at vertex g , let λ_{ig} and $h_{ig} = 1 - \lambda_{ig}$ be its original marginal quantities from Lemma 4.1. Set $h_{ig} = 0$ and $\lambda_{ig} = 1$ for an absent port. The resource contributes no weights. Thus $\sum_g h_{ig} \leq C * s$ for each input source. A gate is *heavy* when $\sum_i h_{ig} > w$. If H gates are heavy, summing their weights gives

$$H \leq \frac{C * sr}{w}. \quad (12)$$

Delete the heavy gates in any fixed order, replacing E_g by P_g at each step. The local difference is

$$D_g P_g D_g - P_g = -2\Pi_g P_g - 2P_g \Pi_g + 4\Pi_g P_g \Pi_g. \quad (13)$$

Each of the three corrections, before its scalar coefficient is restored, factors over the ports of g . Its factor on a present port is of the form ΠP , $P \Pi$, or $\Pi P \Pi$, and hence has rank at most one and operator norm at most one. Trace the wires of g first. For every incident input source, Lemma 4.2 bounds the resulting operator's trace norm by λ_{ig} . A nonincident source is unchanged and has trace norm $1 = \lambda_{ig}$.

The resource contributes at most one, even if it is internally entangled. Indeed, for any operator X on a finite-dimensional register $B \otimes C$ and any contraction T on B , trace-norm duality gives

$$\|\text{Tr}_B[(T \otimes I)X]\|_1 = \sup_{\|R\| \leq 1} |\text{Tr}[(T \otimes R)X]| \leq \|X\|_1. \quad (14)$$

Here R ranges over operators on C ; neither X nor T needs to be Hermitian. Apply this to the resource state and the correction on its port, or leave the resource unchanged if that port is absent. Tensor-independence of the original sources implies that, after this partial trace, the resulting operator is still a tensor product over the sources and resource. Its trace norm is at most $\prod_i \lambda_{ig}$. All other vertex effects act on the complementary wires; their tensor product has operator norm at most one, including effects of previously deleted gates. Duality therefore bounds the absolute contribution of each coefficient-free correction by $\prod_i \lambda_{ig}$. Restoring the coefficients in (13), a deletion costs at most

$$(2 + 2 + 4) \prod_i \lambda_{ig} \leq 8e^{-\sum_i h_{ig}} \leq 8e^{-w},$$

where $1 - h \leq e^{-h}$ was used. Let α^- be the full coefficient after all heavy gates have been deleted. Telescoping and (12) give

$$|\alpha - \alpha^-| \leq 8H e^{-w} \leq 8 \frac{C * sr}{w} e^{-w}.$$

Every step uses the same original F_i and marginal weights. No source is conditioned or renormalized, and no products of single-port penalties from different deletions are asserted.

Refactorization and random activation. Split each deleted gate's base effect P_g into private vertices, one for each of its original ports, including any resource port. This is an exact refactorization of the effect, not a change to any source or to its port partition. Let $\mathcal{V}^-$ be the resulting vertex set. Carry each original port weight to its new vertex, denoting these reindexed weights by h_{iv} ; absent ports again have weight zero. Each port belongs to exactly one vertex, and distinct ports of the same source belong to distinct vertices. The row sums are unchanged. Remaining gate vertices have column weight at most w , input-source private vertices have weight at most $1/2 \leq w$, and resource-only vertices have weight zero.

Write E_v for the positive contraction at vertex $v \in \mathcal{V}^-$. Set $A_v = 2E_v - I_v$, so A_v is Hermitian and $\|A_v\| \leq 1$. Independently include each vertex in a random set $A \subseteq \mathcal{V}^-$ with probability $1/2$. Since $E_v = (I_v + A_v)/2$, the exact activation expansion is

$$\bigotimes_{v \in \mathcal{V}^-} E_v = \mathbb{E}_A \left[\bigotimes_{v \in A} A_v \otimes \bigotimes_{v \notin A} I_v \right].$$

For a fixed A , let $S_i(A)$ be the ports of source i belonging to active vertices. Trace all inactive wires first. The resulting signed operator factors over the input sources and the resource. The resource's active marginal is a density operator with trace norm one, including the scalar one when all its wires are inactive; we do not factor it over its ports. The active vertex operators may couple different sources, but their tensor product still has operator norm at most one. Trace-norm multiplicativity, duality, and the triangle inequality therefore yield

$$|\alpha^-| \leq \mathbb{E}_A \prod_{i=1}^r D_i(A), \quad D_i(A) = \|\text{Tr}_{\mathcal{J}_i \setminus S_i(A)} F_i\|_1, \quad (15)$$

where $\mathcal{J}_i$ is source i 's port index set.

Concentration with shared vertex coins. Let f_i be the majorant from Lemma 4.1 for the original ports of source i , and put

$$F_*(A) = \sum_{i=1}^r f_i(S_i(A)).$$

Distinct ports within a source have distinct vertex coins, so their retention decisions are independent and fair. Consequently $\mathbb{E}_A f_i(S_i(A)) \leq 3/4$, and linearity gives $\mathbb{E}_A F_* \leq 3r/4$. Different sources can share a vertex coin; the random variables $f_i(S_i(A))$ are *not* assumed independent.

Toggling vertex v changes at most one port of each source, and hence changes F_* by at most $c_v = 4 \sum_i h_{iv}$. The column bounds and the unchanged row budgets imply

$$\max_v c_v \leq 4w, \quad \sum_v c_v \leq 4C * sr, \quad \sum_v c_v^2 \leq (\max_v c_v) \sum_v c_v \leq 16C * srw.$$

Since $r \geq 1$ and every source has an owner of weight $1/2$, $\sum_v c_v^2 > 0$. Apply the bounded-differences inequality [22], proved in Appendix A, to the independent vertex coins. Since $7r/8 - \mathbb{E}F_* \geq r/8$, it gives

$$\mathbb{P}[F_* \geq 7r/8] \leq \exp\left(-\frac{2(r/8)^2}{16C * srw}\right) = e^{-r/(512C * srw)}.$$

For every A , the retention majorant and AM–GM give

$$\prod_{i=1}^r D_i(A) \leq \left(\prod_{i=1}^r f_i(S_i(A)) \right)^{1/2} \leq \left(\frac{F_*(A)}{r} \right)^{r/2}.$$

Outside the exceptional event this is at most $(7/8)^{r/2}$; on the event it is at most one since every $D_i \leq 1$. Substituting in (15) yields

$$|\alpha^-| \leq (7/8)^{r/2} + e^{-r/(512C*sw)}.$$

Adding the deletion cost proves (11). Finally, $w = \sqrt{r} \geq 1$ gives exactly $\Delta(r)$ as defined in (3). $\square$

6 Pulling back one output and completing the bound

We first express the measured sign observable as a short linear combination of positive product effects on the two-layer prefix. This local identity does not require first-layer cleanup; only the subsequent application of Theorem 5.1 does. We then combine the cleaned-up estimate with the restriction identity to prove the short quantitative bound.

Lemma 6.1 (One-output coefficient cost). *Let $r \geq 1$ be an integer, and let C be a circuit in the model of Section 2 with r variable inputs. Suppose every first-layer component, as defined in Section 3, contains at most one variable input. Then*

$$|\beta(C)| \leq 6\Delta(r).$$

Empty multiqubit layers and any designated output wire are allowed.

Proof. Local pullback and positive product effects. Work in the product-reflection representation. Let q be the measured wire and $A = M_0 - M_1$ its sign observable. In the notation $U_C = V_3\mathcal{L}_3V_2\mathcal{L}_2V_1\mathcal{L}_1V_0$ of Section 2, first absorb the last one-qubit layer by putting

$$\tilde{A} = V_{3,q}^* A V_{3,q}.$$

This is again a Hermitian contraction; no tracelessness or projective-measurement assumption is made. All gates of $\mathcal{L}_3$ that do not meet q commute with its observable and cancel in the conjugation. Disjointness implies that at most one gate remains.

If that gate is present, let S be its support and write

$$P = \bigotimes_{j \in S \setminus \{q\}} |v_j\rangle\langle v_j|, \quad R = I_q - 2|v_q\rangle\langle v_q|.$$

On S , with q ordered last, the gate is

$$D = I_S - 2P \otimes |v_q\rangle\langle v_q| = (I - P) \otimes I_q + P \otimes R.$$

Since $P^2 = P$ and $P(I - P) = 0$, direct multiplication gives

$$\begin{aligned} D(I \otimes \tilde{A})D &= (I - P) \otimes \tilde{A} + P \otimes R\tilde{A}R \\ &= I \otimes \tilde{A} - 2P \otimes B, \quad B = \frac{\tilde{A} - R\tilde{A}R}{2}. \end{aligned} \tag{16}$$

Identities on wires outside S are understood. This calculation does not depend on whether q was a control or a target before reflection normalization.

The operator B is Hermitian, and $\|B\| \leq (\|\tilde{A}\| + \|R\tilde{A}R\|)/2 \leq 1$. Choose spectral decompositions

$$\tilde{A} = \sum_{a=0}^1 \mu_a Q_a, \quad B = \sum_{b=0}^1 \nu_b T_b, \quad |\mu_a|, |\nu_b| \leq 1,$$

where each of the families $\{Q_0, Q_1\}$ and $\{T_0, T_1\}$ resolves I_q into rank-one orthogonal projectors. Zero eigenvalues are allowed, and a basis in any degenerate eigenspace may be chosen arbitrarily. Substituting into (16) expresses the pulled-back observable as a linear combination of the positive physical-qubit product effects $I \otimes Q_a$ and $P \otimes T_b$, with identities on all other wires. The sum of the absolute coefficients is at most

$$\sum_{a=0}^1 |\mu_a| + 2 \sum_{b=0}^1 |\nu_b| \leq 2 + 4 = 6.$$

Every physical-qubit factor is a positive contraction. If no third-layer gate meets q , the spectral expansion of $\tilde{A}$ alone gives the same conclusion with coefficient sum at most two.

Denote the resulting coefficients and effects by c_ℓ, E_ℓ . To reach the two-layer prefix, also pull back through V_2 . For each product effect E_ℓ , the conjugate $E'_\ell = V_2^* E_\ell V_2$ is still a positive physical-qubit product effect, with unchanged scalar coefficient c_ℓ . Writing $U_{\leq 2} = \mathcal{L}_2 V_1 \mathcal{L}_1 V_0$ and ρ_x^{in} for the initial state, we have the exact identity

$$a_C(x) = \sum_{\ell} c_\ell \text{Tr}[E' * \ell U * \leq 2 \rho_x^{\text{in}} U_{\leq 2}^*], \quad \sum_{\ell} |c_\ell| \leq 6.$$

The coefficients and effects are independent of x . Everything up to this point holds without a restriction on first-layer input arities, as needed for the later exponential argument.

Applying the cleaned-up source estimate. First assume pure product ancillas, using the reduction in Section 2. Each input-containing first-layer component then supplies one independent source of Section 4. Group all components containing no variable input into a single input-independent resource density operator. It is tensor-independent of those sources, though it may be internally entangled. The local layer V_1 preserves this factorization. Each E'_ℓ factors positively over all source and resource ports of $\mathcal{L}_2$, so Theorem 5.1 bounds its full parity coefficient by $\Delta(r)$. Linearity and the triangle inequality give $|\beta(C)| \leq \sum_{\ell} |c_\ell| \Delta(r) \leq 6\Delta(r)$. Finally, the input-independent convex decomposition of product mixed ancillas from Section 2 extends this uniform bound to the stated model. $\square$

Proof of Theorem 2.2. Restriction and the finite-parameter bound. Fix an integer k with $2 \leq k \leq n+1$. Delete the first-layer gates containing at least k original inputs and choose the survivor set J as in Section 3. This set is independent of the input assignment. Its size satisfies $r = |J| \geq \lceil n/(k-1) \rceil \geq 1$. For every fixed assignment z to $[n] \setminus J$, the restricted circuit C_z^- satisfies Lemma 6.1, including when the designated output wire belongs to the fixed inputs. Thus $|\beta(C_z^-)| \leq 6\Delta(r)$ for every z . The exact identity (7) gives

$$|\beta(C^-)| = |\mathbb{E}_z[\chi_{[n] \setminus J}(z) \beta(C_z^-)]| \leq \mathbb{E}_z |\beta(C_z^-)| \leq 6\Delta(r).$$

This also covers $J = [n]$, when the sole complementary assignment is empty and its parity sign is one.

The function Δ is decreasing on its real domain $[1, \infty)$. Indeed, writing $y = \sqrt{r} \geq 1$, the first term is $8C * \text{sy}e^{-y}$ and $(ye^{-y})' = e^{-y}(1-y) \leq 0$; the other two terms decrease directly. Since $k \leq n+1$, one has $n/(k-1) \geq 1$, so $\Delta(r) \leq \Delta(n/(k-1))$ is within this domain. The deletion estimate of Section 3 and the triangle inequality now give

$$\begin{aligned} |\beta(C)| &\leq |\beta(C) - \beta(C^-)| + |\beta(C^-)| \\ &\leq 4\sqrt{\frac{n}{k}} 2^{-k} + 6\Delta(r) \leq 4\sqrt{\frac{n}{k}} 2^{-k} + 6\Delta\left(\frac{n}{k-1}\right), \end{aligned}$$

which proves (4) for every admissible k .

The asymptotic bound, with explicit constants. For $n \geq 8$, put $t = n^{1/3} \geq 2$ and $k = \lceil t \rceil$. This is admissible: $2 \leq k \leq n + 1$. Moreover, $k \geq t$, $0 < k - 1 < t$, and hence

$$\sqrt{\frac{n}{k}} \leq t, \quad 2^{-k/2} \leq e^{-(\log 2)t/2}, \quad \frac{n}{k-1} \geq t^2.$$

Monotonicity therefore gives $\Delta(n/(k-1)) \leq \Delta(t^2)$. Consequently

$$\begin{aligned} |\beta(C)| &\leq 4te^{-(\log 2)t/2} + 48C * ste^{-t} \\ &\quad + 6e^{-(\log(8/7))t^2/2} + 6e^{-t/(512C*s)}. \end{aligned} \tag{17}$$

To make the absorption of the prefactors explicit, set

$$a = \frac{\log 2}{2}, \quad c = \frac{1}{1024C * s}.$$

Then $0 < c < \min\{a, 1, \log(8/7)\}$: for example, $C * s \geq 1/2$, $\log 2 \geq 1/2$, and $\log(8/7) \geq 1/8$ suffice. For $d > 0$, $\sup_{s \geq 0} se^{-ds} = 1/(ed)$, so

$$4te^{-at} \leq \frac{4}{e(a-c)}e^{-ct}, \quad 48C * ste^{-t} \leq \frac{48C * s}{e(1-c)}e^{-ct}.$$

For $t \geq 2$, the quadratic term in (17) is at most $6e^{-ct}$ because $(\log(8/7))t^2/2 \geq (\log(8/7))t \geq ct$. Its last term is $6e^{-2ct} \leq 6e^{-ct}$. Thus one may take

$$K_* = \frac{4}{e(a-c)} + \frac{48C * s}{e(1-c)} + 12, \quad K = \max\{K_*, e^{2c}\}.$$

For $1 \leq n < 8$ one has $n^{1/3} < 2$, and the elementary bound $|\beta(C)| \leq 1 \leq e^{2c}e^{-cn^{1/3}}$ covers those cases. This proves (5) with absolute constants independent of all circuit parameters specified in the theorem. $\square$

7 An explicit average-correlation construction

The following reversible classical circuits are also circuits in our quantum model. They give a nonzero exponential-scale correlation with parity, as well as exact small-input examples. No preservation of the other input wires is required; they may retain garbage, as permitted in Section 2.

Proposition 7.1. *For every integer $n \geq 1$, put $r = \lceil n/4 \rceil$. There is a zero-ancilla circuit C_n on the n input wires, with at most three multiqubit layers and at most $n - r + 1$ multiqubit gates, such that*

$$\beta(C_n) = \begin{cases} 1, & r = 1, \\ 2^{2-r}, & r \geq 2. \end{cases}$$

In particular, for $1 \leq n \leq 8$ the designated output equals parity on every input.

Proof. Two-layer block preprocessing. Partition the inputs, independently of their values, into nonempty blocks $G_1, \dots, G_r$ of size at most four; for example, take consecutive blocks of four and, if needed, a smaller final block. Within a block of size b , label its wires $q_1, \dots, q_b$ and choose q_b as its

representative. In the following schedule, $i \rightarrow j$ denotes a CNOT with control q_i and target q_j , and a dash denotes no gate:

b	first layer	second layer
1	—	—
2	$1 \rightarrow 2$	—
3	$1 \rightarrow 2$	$2 \rightarrow 3$
4	$1 \rightarrow 2, 3 \rightarrow 4$	$2 \rightarrow 4$

For $b = 3$, the second gate adds the parity of the first two inputs into the third. For $b = 4$, it adds the parity of the first pair into that of the second pair. The cases $b = 1, 2$ are immediate. Thus the representative of block G_j holds

$$z_j = \bigoplus_{i \in G_j} x_i.$$

Within each layer all gates have disjoint supports, including their controls; for $b = 4$, the two first-layer gates use the disjoint pairs $\{q_1, q_2\}$ and $\{q_3, q_4\}$. Different blocks use disjoint wires, so all block schedules run in at most two global layers. Each block uses exactly $b - 1$ CNOTs, for a total of $\sum_j (|G_j| - 1) = n - r$. All gates act in place on input wires; no ancillary qubit, measurement, or extra input copy is used in this preprocessing.

If $r = 1$, omit the final multiqubit gate and measure the sole representative in the computational basis. It already holds the full parity. This uses $n - 1$ CNOTs and at most two layers, including no multiqubit gate when $n = 1$.

Final gate and exact correlation for $r \geq 2$. For a uniform original input, each block parity is uniform and the block parities are independent. More explicitly, every $z \in \{0, 1\}^r$ has exactly $\prod_j 2^{|G_j|-1} = 2^{n-r}$ original input preimages. Moreover $\bigoplus_j z_j = \bigoplus_i x_i$.

Apply one generalized Toffoli in the third layer, with controls on the first $r - 1$ representatives and target on the last one. Its target bit is

$$y_0 = z_r \oplus a(z), \quad a(z) = \prod_{j=1}^{r-1} z_j.$$

There is just one gate in this layer. For $r = 2$ it is a CNOT. Before any output NOT, the parity correlation is

$$\begin{aligned} \mathbb{E}_x(-1)^{\sum_i x_i + y_0(x)} &= \mathbb{E}_z(-1)^{\sum_{j=1}^{r-1} z_j + a(z)} \\ &= \mathbb{E}_z(-1)^{\sum_{j=1}^{r-1} z_j} (1 - 2a(z)) \\ &= -2(-1)^{r-1} 2^{-(r-1)} = (-1)^r 2^{2-r}. \end{aligned} \tag{18}$$

Here the contribution without $a(z)$ is zero because $r - 1 \geq 1$ and the control bits are independent and uniform. The remaining contribution is supported only on the all-ones control string, which has probability $2^{-(r-1)}$. The target bit z_r cancels from the parity sign, so there is no additional factor of $1/2$.

For odd $r \geq 3$, apply a NOT to the target after the final gate; if r is even, do not. This input-independent choice makes the correlation 2^{2-r} positive. The NOT belongs to a local one-qubit layer and is not counted as a multiqubit gate or an additional multiqubit layer. Measure only this target wire in the computational basis; no classical processing of other output wires is needed. The total multiqubit gate count is exactly $n - r + 1$ for $r \geq 2$.

Finally, when $r = 2$, no output NOT is used and $y_0 = z_1 \oplus z_2$ is the full parity on every input. Together with the $r = 1$ case, this proves the exact assertion for $1 \leq n \leq 8$, as well as the stated count and correlation. $\square$

Average advantage and worst-case behavior. Let X be uniform on $\{0, 1\}^n$, and write $C_n(x)$ for the deterministic measured output bit on input x . For $r \geq 2$, the uniform-input success probability and advantage are, respectively,

$$\mathbb{P}[C_n(X) = \text{PARITY}_n(X)] = \frac{1}{2} + 2^{1-r}, \quad \mathbb{P}[C_n(X) = \text{PARITY}_n(X)] - \frac{1}{2} = 2^{1-r}.$$

For $n \geq 9$, equivalently $r \geq 3$, this is a positive but exponentially small *average* advantage. These particular circuits are deterministic and do not compute parity on every input: for odd $r \geq 3$, all-zero controls yield an incorrect answer after the output NOT; for even $r \geq 4$, a control string with exactly one one yields an incorrect answer. Every such control pattern has original input preimages. Consequently their worst-case success probability is zero. This statement does not apply to the exact $n \leq 8$ cases and does not assert an impossibility for other circuits at $n = 9$ or any fixed input length.

As $n \rightarrow \infty$, the displayed correlation satisfies

$$\beta(C_n) = \exp\left(-\frac{\log 2}{4}n + O(1)\right).$$

Thus nonzero parity correlation is achievable with zero ancillas and linearly many gates, matching the exponential-in- n order of Theorem 2.1. No optimal exponential constant or sharp threshold for exact depth-three computation is asserted. In particular, the exact small-input examples are compatible with the asymptotic lower bound.

A The bounded-differences inequality

We record both the tail and exponential-moment forms of the bounded-differences inequality [22]. The proof uses the centered bounded-variable estimate of Hoeffding [23] and is included to specify the constants used in the appendix arguments.

Lemma A.1 (Bounded differences and exponential moments). *Let $m \geq 0$ be an integer, and let $X_1, \dots, X_m$ be independent random variables with finite supports $\mathcal{X}_1, \dots, \mathcal{X}_m$. Let $F : \prod_{j=1}^m \mathcal{X}_j \rightarrow \mathbb{R}$ satisfy $|F(x) - F(x')| \leq c_j$ whenever x, x' differ only in coordinate j , where $c_j \geq 0$. Put $V = \sum_{j=1}^m c_j^2$. Then, for every real t ,*

$$\mathbb{E}e^{t(F - \mathbb{E}F)} \leq e^{t^2 V / 8}. \quad (19)$$

If $V > 0$, then for every $u > 0$,

$$\mathbb{P}[F - \mathbb{E}F \geq u] \leq \exp\left(-\frac{2u^2}{V}\right).$$

If $V = 0$, then F is constant on the product support and this upper-tail probability is zero. These statements include $m = 0$, when the product support is a singleton and $V = 0$.

Proof. First let Y be a centered finite random variable in an interval of length c . Its log moment generating function $\phi(t) = \log \mathbb{E}e^{tY}$ satisfies $\phi(0) = \phi'(0) = 0$. Differentiating the finite sum gives $\phi''(t) = \text{Var}_t(Y)$, where the subscript denotes the probability distribution tilted by the positive factor e^{tY} . This distribution has the same interval support. If a is the interval midpoint, then $\text{Var}_t(Y) \leq \mathbb{E}_t(Y - a)^2 \leq c^2/4$. Integrating twice gives $\phi(t) \leq t^2 c^2 / 8$ for $t \geq 0$; applying this to $-Y$ gives the same bound for $t < 0$.

Now set $M_j = \mathbb{E}[F \mid X_1, \dots, X_j]$ for $0 \leq j \leq m$, so $M_0 = \mathbb{E}F$ and $M_m = F$. Conditional on any preceding values of positive probability, M_j is a function of X_j whose range has length at most

c_j : independence gives the same law of the future coordinates for every value of X_j , and averaging preserves the assumed coordinate oscillation. Thus $Y_j = M_j - M_{j-1}$ is conditionally centered and lies in an interval of length at most c_j . The preceding estimate gives

$$\mathbb{E}[e^{tY_j} \mid X_1, \dots, X_{j-1}] \leq e^{t^2 c_j^2 / 8}.$$

Iterating conditional expectations and using $\sum_j Y_j = F - \mathbb{E}F$ proves (19). For $V > 0$, exponential Markov gives $\mathbb{P}[F - \mathbb{E}F \geq u] \leq \exp(-tu + t^2 V / 8)$ for $t > 0$; choose $t = 4u/V$. If $V = 0$, all coordinate oscillations vanish. Any two points of the product support can be joined by successive single-coordinate changes, so F is constant there. The empty product case has the same conclusion directly. $\square$

B Exponential correlation decay

This appendix strengthens the correlation rate to exponential in n . It is not needed for the size-independent bounded-error conclusion already obtained in the main text. Its additional ingredient is control of source retention after several heavy-gate corrections have been inserted. All inserted operators remain unnormalized.

B.1 Retention after a joint rank-one insertion

Lemma B.1 (Joint insertion and elementary trace bound). *Let F be a one-input source from Section 4, with port index set $\mathcal{J}$, and let $J \subseteq \mathcal{J}$. Identify J with the union of its port registers when writing partial traces or operators on it. For a rank-at-most-one contraction T on that union, put*

$$K = \text{Tr}_J[(T \otimes I)F], \quad \nu = \|K\|_1, \quad \tau = \text{Tr } K.$$

Then

$$\nu \leq \lambda_{\max}(\bar{\rho}_J) \leq 1, \quad |\tau| \leq \frac{1}{2}.$$

For the empty union use the scalar $T = 1$, so $K = F$ and $\tau = 0$.

Proof. The proof of Lemma 4.2 applies unchanged when the entire union is treated as one tensor factor. For the trace bound, $F_J = \text{Tr}_{J^c} F$ is Hermitian, traceless, and has trace norm at most one by (14). Its positive and negative parts therefore each have trace at most one half, so $\|F_J\| \leq 1/2$. Since T has rank at most one and norm at most one, $\|T\|_1 \leq 1$. Hence $|\tau| = |\text{Tr}(TF_J)| \leq \|T\|_1 \|F_J\| \leq 1/2$. The argument includes complex, non-Hermitian T . $\square$

Lemma B.2 (Post-insertion retention). *With the notation of Lemma B.1, let $\mathcal{R} = \mathcal{J} \setminus J$ be the remaining ports and keep their original weights h_j . There is $f : 2^{\mathcal{R}} \rightarrow [0, \nu^2]$ such that*

$$\|\text{Tr}_{\mathcal{R} \setminus A} K\|_1^2 \leq f(A), \quad \mathbb{E}_A f \leq \frac{13}{16}, \quad |f(A \triangle \{j\}) - f(A)| \leq 4h_j.$$

The expectation uses independent fair retention of the remaining ports, and the oscillation bound holds for every $A \subseteq \mathcal{R}$ and $j \in \mathcal{R}$. If the owner is not inserted, one also has

$$\mathbb{E}_A f \leq \frac{3\nu^2 + |\tau|^2}{4}. \tag{20}$$

Proof. If $T = 0$ take $f = 0$. If the owner is inserted, its joint averaged marginal has a pure data-zero block of weight one half and an orthogonal block of total weight one half, as in (8). Thus $\nu \leq 1/2$, and the constant majorant $f = \nu^2 \leq 1/4$ suffices by partial-trace contractivity. It also covers an empty remaining register.

Suppose the owner remains. Undo final source-local unitaries within each port; absorb the inserted-register factors into T . This preserves all relevant norms and weights. Write $T = |u\rangle\langle v|$ with $\|u\|, \|v\| \leq 1$. Let β and γ be the initial product vectors on the inserted and remaining ancillary registers, respectively; γ includes any owner ancillas. Put

$$P_B = \langle \beta | \Pi_B | \beta \rangle, \quad P_R = \langle \gamma | \Pi_R | \gamma \rangle, \quad P = P_B P_R.$$

Here Π_B and Π_R are the restrictions of the original ancillary trigger projector to the inserted and remaining ancillary registers. Empty ancillary registers have unit scalar vector and projector. The data wire is not part of either of these two registers. If $P = 0$, the source gate acts identically on its logical input subspace. The remaining K is a scalar times the original traceless signed data operator tensored with a pure ancillary state. Take $f = \nu^2$ when the owner is retained and zero otherwise. Its mean is $\nu^2/2$, ancillary toggles do nothing, and an owner toggle costs at most $1 \leq 4h_0$. Here $\tau = 0$, so the mean also satisfies (20), since $\nu^2/2 \leq 3\nu^2/4$.

Now assume $P > 0$. Choose a unit vector η with $\Pi_R \gamma = \sqrt{P_R} \eta$. For $z = u, v$ define

$$t_z = \langle z | \beta \rangle, \quad b_z = \langle z | \Pi_B | \beta \rangle, \quad |t_z| \leq 1, \quad |b_z| \leq \sqrt{P_B}.$$

In the source's data-trigger basis its signed input is $S/2$, where

$$S = \begin{pmatrix} \xi & \zeta \\ \bar{\zeta} & -\xi \end{pmatrix}, \quad \xi \in \mathbb{R}, \quad \xi^2 + |\zeta|^2 = 1.$$

Define compressed maps by

$$W_z |0\rangle = |0\rangle \otimes t_z \gamma, \quad W_z |1\rangle = |1\rangle \otimes (t_z \gamma - 2b_z \sqrt{P_R} \eta).$$

Direct partial tracing gives the order-sensitive identity $K = W_v S W_u^* / 2$.

Erase the data and any further ancillary wires, retaining an ancillary register C . Let Q be the product of the erased ancillary trigger probabilities, so $P_R \leq Q \leq 1$. Set

$$x = t_v \bar{b}_u, \quad y = b_v \bar{t}_u, \quad z = b_v \bar{b}_u, \quad L = |x| + |y|, \quad \tau_0 = P_R(x + y - 2z).$$

Subtracting the two data diagonal blocks gives

$$K_C = \xi Q (x \rho_C \Pi_C + y \Pi_C \rho_C - 2z \Pi_C \rho_C \Pi_C).$$

Here $\rho_C = |\gamma_C\rangle\langle\gamma_C|$ is the initial pure product state on C , and Π_C is the restriction of the ancillary trigger projector to C . For $C = \emptyset$ both are the scalar one. In the trigger/nontrigger orthonormal basis of the span of the retained initial vector and its trigger vector, this is

$$K_C = \xi \begin{pmatrix} \tau_0 & y \sqrt{P_R(Q - P_R)} \\ x \sqrt{P_R(Q - P_R)} & 0 \end{pmatrix}.$$

If the retained trigger probability is one, the off-diagonal entries vanish and the same formula is interpreted on its one-dimensional support; this includes the empty register. Because $\|M\|_1^2 = \text{Tr}(M^* M) + 2|\det M|$ for a two-by-two matrix,

$$\|K_C\|_1^2 = |\tau|^2 + \xi^2 L^2 P_R(Q - P_R). \quad (21)$$

Indeed, $\text{Tr } K_C = \text{Tr } K = \tau = \xi\tau_0$, and the two off-diagonal entries contribute $\xi^2(|x|+|y|)^2 P_R(Q-P_R)$ to the squared trace norm. This is a singular-value identity, not a Hermitian eigenvalue calculation: x and y need not be conjugates.

Take $f = \nu^2$ when the owner is retained, and take the exact squared retained norm otherwise. Partial-trace contractivity gives $f \leq \nu^2$. Conditional on owner erasure, let p_0 be the owner-ancilla trigger probability and let p_j range over other remaining ports. The product inequality from Lemma 4.1 gives

$$\mathbb{E}Q = p_0 \prod_j \frac{1+p_j}{2} \leq \frac{p_0 + P_R}{2} \leq \frac{1 + P_R}{2}.$$

The auxiliary partial trace that erases only the data has $Q = 1$ and norm at most ν , even if it retains ancillas in the owner. Thus (21) implies

$$|\tau|^2 + \xi^2 L^2 P_R(1 - P_R) \leq \nu^2.$$

The owner-erased mean is at most $(\nu^2 + |\tau|^2)/2$. Averaging the owner's independent fair coin proves (20). Lemma B.1 then bounds it by $(3 + 1/4)/4 = 13/16$.

For an ancillary port j changing from erased to retained while the owner is erased, (21) changes by

$$\xi^2 L^2 P_R Q_{\text{rest}}(1 - p_j) \leq 4P_B P_R(1 - p_j) = 4P(1 - p_j) \leq 4h_j.$$

Here $L \leq 2\sqrt{P_B}$ and $Q_{\text{rest}} \leq 1$. The last weight is the original one from (9); it is not recomputed on K . Ancillary toggles do nothing when the owner is retained, and an owner toggle costs at most $\nu^2 \leq 1 \leq 4h_0$. $\square$

B.2 Summing all heavy-gate corrections

Theorem B.3 (Exponential source-event bound). *There are absolute constants $K_0 \geq 1$ and $a > 0$ such that a positive product-of-port event in the two-layer source network of Theorem 5.1 has full coefficient at most $K_0 e^{-ar}$ in magnitude. Here $r \geq 0$ is an integer, the event factors are positive contractions, and the independent resource may be any complex operator σ with $\|\sigma\|_1 \leq 1$. The functional need not be a probability when σ is not positive.*

Proof. For $r = 0$, trace-norm duality gives magnitude at most one. Assume $r \geq 1$ and use the original port weights. Fix $w \geq 1$ temporarily, and call a real gate heavy if its column load exceeds w . Let $\mathcal{H}$ be their set and $H = |\mathcal{H}|$, so $H \leq C * sr/w$.

Instead of telescoping deletions, expand every heavy gate by (13), including its base term P_g . A branch chooses a set $S \subseteq \mathcal{H}$ of correction gates and one of three correction types at each selected gate. With $k = |S|$, the sum of the absolute coefficients over the types is 8^k , not $3^k 8^k$. For source i , tensor its selected rank-one port contractions into *one* rank-at-most-one contraction T_i on the joint selected union J_i . Insert it once to obtain $K_i = \text{Tr}_{J_i}[(T_i \otimes I)F_i]$. The resource remainder has norm at most one by (14). Unselected heavy base terms split into private positive port effects. All remaining real gates are light. Every surviving effect is a positive contraction, and all remaining ports retain their original weights.

For the coefficient-free branch $\mathcal{T}_{S,\ell}$, the exact fair-vertex expansion used in (15) gives

$$|\mathcal{T}_{S,\ell}| \leq \mathbb{E}_A \prod_i D_i(A), \quad D_i(A) = \|\text{Tr}_{\text{inactive remaining ports}} K_i\|_1.$$

The complex resource is kept jointly; every further partial trace still has norm at most one. Lemma B.2 gives majorants with mean at most 13/16 and original-weight toggles. Distinct remaining ports

within a source still belong to distinct vertices, including the private vertices created from unselected heavy base terms. Their coins are therefore independent and fair. Different sources may share a vertex coin; independence between their retained norms is not used. Thus, for $F_* = \sum_i f_i$,

$$\mathbb{E}F_* \leq 13r/16, \quad \sum_g c_g^2 \leq 16C * srw.$$

At threshold $7r/8$ the upward deviation is at least $r/16$. Bounded differences and AM–GM therefore give the uniform bound

$$|\mathcal{T}_{S,\ell}| \leq (7/8)^{r/2} + e^{-r/(2048C*sw)} \leq 2e^{-c_0 r/w}, \quad c_0 = (2048C * s)^{-1}. \quad (22)$$

If all toggle constants vanish, F_* is constant below the threshold, so the exceptional probability is zero. The last inequality uses $w \geq 1$ and $\frac{1}{2} \log(8/7) > c_0$.

A second bound penalizes dense correction sets. For each source write

$$W_i(S) = \sum_{g \in S} h_{ig}, \quad H_i(S) = 1 - \lambda_{\max}(\bar{\rho}_{i,J_i}).$$

The empty union has both quantities zero. Joint insertion gives $\|K_i\|_1 \leq 1 - H_i(S)$. If J_i contains the owner, $H_i(S) = 1/2 \geq W_i(S)/(2C * s)$. Otherwise let P_i be the full original ancillary trigger probability and let p_{J_i} be its product on the selected union. Treating that union as one ancillary port in (9) gives, when $P_i > 0$,

$$H_i(S) \geq P_i(1 - p_{J_i}), \quad W_i(S) \leq 2P_i \log(1/p_{J_i}), \quad p_{J_i} \geq P_i.$$

If $P_i = 0$, all ancillary-only weights vanish.

Fix $0 < \delta < e^{-1}$. The function $(1 - p)/\log(1/p)$ is increasing on $(0, 1)$: its derivative has the sign of $\log p + p^{-1} - 1 \geq 0$. Thus for $P_i \geq \delta$, and also for any owner-containing union,

$$H_i(S) \geq c_\delta W_i(S), \quad c_\delta = \min \left\{ \frac{1}{2C * s}, \frac{1 - \delta}{2 \log(1/\delta)} \right\} > 0.$$

The case $p_{J_i} = 1$ has zero ancillary weight and follows directly. Discarding only owner-free rows with $P_i < \delta$ loses at most $2r\delta \log(1/\delta)$ of selected weight, because $P \log(1/P)$ is increasing on $(0, e^{-1})$. Selected heavy columns have total load at least kw . Hence

$$\sum_i H_i(S) \geq c_\delta (kw - 2r\delta \log(1/\delta)).$$

All remaining effects are contractions. Tensor-norm duality gives

$$|\mathcal{T}_{S,\ell}| \leq \prod_i \|K_i\|_1 \leq \exp[-c_\delta (kw - 2r\delta \log(1/\delta))]. \quad (23)$$

This uses one joint penalty per source. It does not multiply individual-port penalties on the same source.

Choose an absolute $\eta \in (0, C * s)$ such that $A_0 = \eta \log(8eC * s/\eta) < c_0$. Then choose $\delta \in (0, e^{-1})$ with $2\delta \log(1/\delta) \leq \eta/2$. Finally choose a fixed $w \geq 1$ such that

$$\frac{8eC * s}{\eta} e^{-c_\delta w/2} \leq e^{-1}.$$

These sequential choices exist and are independent of the circuit. Put $u = r/w$ and $b = c_0 - A_0 > 0$.

For integers $1 \leq k \leq \min\{H, \eta u\}$,

$$\binom{H}{k} 8^k \leq \exp(k \log(8eC * su/k)) \leq e^{A_0 u}.$$

The final step uses monotonicity of $s \log(8eC * su/s)$ on $0 < s \leq \eta u$. The $k = 0$ term is one, and there are at most $C * su + 1$ possible values of k . Multiplying by (22), all sparse correction sets contribute at most $2(C * su + 1)e^{-bu}$.

For $k > \eta u$, one has $2r\delta \log(1/\delta) \leq \eta r/2 < kw/2$. Using (23), the full contribution of size k is at most

$$\binom{H}{k} 8^k e^{-c_\delta kw/2} \leq \left(\frac{8eC * s}{\eta} e^{-c_\delta w/2} \right)^k \leq e^{-k}.$$

Summing the geometric series over integers $k > \eta u$ bounds the dense part by $e^{-\eta u}/(1 - e^{-1})$. This includes empty ranges, $H = 0$, and $\eta u < 1$.

Choose

$$a = \frac{\min\{b/2, \eta\}}{w}, \quad K_0 = 2 + \frac{4C * s}{eb} + \frac{1}{1 - e^{-1}}.$$

Using $\sup_{u \geq 0} ue^{-bu/2} = 2/(eb)$, the sparse and dense bounds sum to at most $K_0 e^{-ar}$. Also $K_0 > 1$, which covers the already verified case $r = 0$. $\square$

B.3 Removing the restriction on first-layer input arities

Theorem B.4 (Arbitrary first-layer inputs). *There are absolute $K_0 \geq 1$ and $\kappa > 0$ such that every physical-qubit product event $E = \bigotimes_q E_q$, with $0 \leq E_q \leq I_q$, after two reflection layers has full n -input parity coefficient of magnitude at most $K_0 e^{-\kappa n}$, with arbitrary finite product ancillary initialization and circuit width. The constants are independent of the event and the circuit.*

Proof. By the input-independent product-state decomposition in Section 2, it suffices to prove a bound uniform over pure product ancillary initializations. Assume this from now on and use the constants in Theorem B.3. Put

$$b = \min\{a, (\log 2)/2\}, \quad L = \max\left\{2, \left\lceil \frac{2 \log(8/b)}{\log 2} \right\rceil\right\}, \quad \kappa = b/L > 0.$$

The source theorem remains valid with the smaller rate b . Call a first-layer gate large if it contains more than L inputs. Let k_g be that number for each large gate.

For a first-layer gate with k inputs, the signed pre-gate operator is

$$F_0 = \bigotimes_{j=1}^k \frac{S_j}{2} \otimes \bigotimes_{q \in \text{anc}} |\alpha_q\rangle\langle\alpha_q|,$$

where each S_j is a traceless Hermitian one-qubit unitary. Its operator norm is 2^{-k} . Write the rank-one gate projector as $\Pi = |p\rangle\langle p|$, with $\|p\| = 1$, and put $v = (I - \Pi)F_0|p\rangle$. The signed post-gate operator has the binary expansion

$$DF_0D = F_0 + \Delta, \quad \Delta = -2(|p\rangle\langle v| + |v\rangle\langle p|).$$

Indeed, expand $DF_0D - F_0 = -2\Pi F_0 - 2F_0\Pi + 4\Pi F_0\Pi$ and cancel its Π -diagonal terms. Since $v \perp p$, the nonzero eigenvalues of Δ are $2\|v\|$ and $-2\|v\|$ when $v \neq 0$. Consequently

$$\|\Delta\|_1 = 4\|v\| \leq 4\|F_0\| = 4 \cdot 2^{-k}.$$

Thus $\Delta = 4 \cdot 2^{-k} \sigma_g$ for a Hermitian resource operator σ_g of trace norm at most one, with $\sigma_g = 0$ if $\Delta = 0$. This resource may be entangled within the component. It is independent of the remaining input variables: all k variables of its component have already been signed and averaged into F_0 . The baseline F_0 deletes the gate and leaves all k variables as independent singleton sources. Its ancillary factors remain pure and product and can be included in the independent resource.

Expand only the large first-layer gates. In every other input-containing component keep one variable and condition the others classically. Include untouched input singletons among these small components. Let r_0 be their number and m their total number of inputs, so $m \leq Lr_0$. For a fixed correction subset S of large gates, the remaining number of variable sources is exactly

$$r(S) = r_0 + \sum_{g \notin S} k_g.$$

Group all corrected components, all input-free components, and the baseline ancillary factors into one resource. After extracting the factors $4 \cdot 2^{-k_g}$, this resource has trace norm at most one and is tensor-independent of all remaining input sources. This remains true when the resource is not positive. Local unitaries after the first layer preserve the norms and all tensor separations used here.

The event factors over physical wires, so it factors over every new source-by-second-gate port partition, including after a large baseline component has split into singleton sources. This physical factorization hypothesis is essential at this interface. Apply Theorem B.3 to every fixed assignment of the conditioned inputs and every correction branch. Restoring their parity signs and averaging cannot increase the absolute bound. Summing the extracted correction factors over subsets therefore gives the generating-function upper bound

$$|\widehat{q}([n])| \leq K_0 e^{-br_0} \prod_{g \text{ large}} \left(e^{-bk_g} + 4 \cdot 2^{-k_g} \right). \quad (24)$$

This also covers branches with no surviving sources, by the $r = 0$ case of the source theorem.

For $k > L$, the chosen cutoff and $b \leq (\log 2)/2$ give

$$8e^{-(\log 2 - b)k} \leq 8 \cdot 2^{-k/2} \leq b \leq bk.$$

Consequently, using $1 + t \leq e^t$,

$$e^{-bk} + 4 \cdot 2^{-k} \leq e^{-bk} (1 + bk/2) \leq e^{-bk/2}.$$

Moreover $e^{-br_0} \leq e^{-bm/L}$ and $b/L \leq b/2$. Since $n = m + \sum_{g \text{ large}} k_g$, equation (24) is at most $K_0 e^{-\kappa n}$. Empty products, no large gates, and no small components are included without a separate normalization or limiting argument. $\square$

Corollary B.5 (Exponential one-output correlation). *Every three-layer circuit in the model of Section 2 satisfies*

$$|\beta(C)| \leq 6K_0 e^{-\kappa n}.$$

Its uniform-input parity advantage is at most $3K_0 e^{-\kappa n}$.

Proof. Use the one-output pullback in Lemma 6.1, whose positive physical-product coefficient mass is at most six, and apply Theorem B.4 to each term. The input sources now need not be cleaned up. Mixed product ancillas follow by convexity as in Section 2. $\square$

Proposition 7.1 gives achievable correlation $\exp(-\Theta(n))$ with no ancillas and linearly many gates. Thus the exponential order of this upper bound is qualitatively tight; no optimal exponential constant is asserted.

B.4 One explicit choice of constants

The constants can be made fully numerical without optimizing them. In Theorem B.3 choose

$$\eta = 2^{-20}, \quad \delta = 2^{-32}, \quad w = 2048.$$

Use $27/10 < e < 3$, $\log 2 < 7/10$, and $C * s < 5/4$. Then $\log(8eC * s/\eta) < 18$, so

$$A_0 < 18 \cdot 2^{-20}, \quad c_0 > 1/2560, \quad c_0 - A_0 > 373/10^6.$$

Also

$$2\delta \log(1/\delta) < 2^{-26} < \eta/2, \quad c_\delta > 1/50.$$

The last inequality follows for both entries of the defining minimum: $1/(2C * s) > 2/5$ and $(1 - 2^{-32})/(64 \log 2) > 1/50$. Thus

$$\log(8eC * s/\eta) - c_\delta w/2 < 18 - 20.48 < -1,$$

which verifies the required dense-sum condition. Moreover $(c_0 - A_0)/2 > \eta$, so the source rate is exactly $a = \eta/w = 2^{-31}$. Its prefactor obeys

$$K_0 < 4 + \frac{5}{(27/10)(373/10^6)} < 5000.$$

We may therefore enlarge it to 5000.

For Theorem B.4, one has $b = a = 2^{-31}$ and $8/b = 2^{34}$, giving the integer cutoff $L = 68$ exactly. Hence $\kappa = 2^{-31}/68 > 2^{-38}$. Corollary B.5 consequently gives the explicit bound

$$\boxed{|\beta(C)| \leq 30000 \exp(-2^{-38}n)}. \quad (25)$$

For small n , the elementary bound $|\beta(C)| \leq 1$ is stronger. The purpose of (25) is to exhibit positive, uniform constants, not a useful numerical threshold.

C An alternative retention proof for arbitrary state pairs

This appendix gives a second proof of the retention property needed by the heavy-set expansion. Unlike Lemma B.2, it does not calculate the source's reduced matrices. It applies to arbitrary mixed state pairs satisfying the owner-weight and total weight hypotheses below, and uses only their original marginal weights. The two arguments share the network expansion and the reflection-specific dense-set penalty; they are not independent proofs of those shared ingredients.

C.1 Purity and a rank-one deficit inequality

Lemma C.1 (Common pure-state overlap). *For density operators ρ_0, ρ_1 , put $\sigma = (\rho_0 + \rho_1)/2$ and $d = \|\rho_0 - \rho_1\|_1/2$. If z is a unit vector and $p = \langle z | \sigma | z \rangle$, then*

$$d^2 \leq 1 - (2p - 1) * +^2, \quad t * + = \max\{t, 0\}.$$

Proof. Set $p_i = \langle z | \rho_i | z \rangle$. On a sufficiently large common ancillary space choose purifications whose z components point in the same ancillary direction, represented by a unit vector e :

$$\Psi_i = \sqrt{p_i} z \otimes e + \xi_i, \quad (\langle z | \otimes I) \xi_i = 0, \quad \|\xi_i\|^2 = 1 - p_i.$$

Independent ancillary unitaries achieve this alignment; a zero component imposes no condition. If $p_0 + p_1 \geq 1$, then

$$|\langle \Psi_0 | \Psi_1 \rangle| \geq \sqrt{p_0 p_1} - \sqrt{(1-p_0)(1-p_1)} \geq p_0 + p_1 - 1.$$

For the last inequality, multiply the difference of square roots by their sum. The product is $p_0 + p_1 - 1$, and the sum is at most one by Cauchy–Schwarz. A zero denominator gives a zero required lower bound. The pure-state trace-distance formula and partial-trace contractivity prove the result. For $p_0 + p_1 < 1$ it is just $d \leq 1$. $\square$

Lemma C.2 (Sharp deficit under a rank-one contraction). *Let ρ_0, ρ_1 be any density operators on $B \otimes A$, and put $F = (\rho_0 - \rho_1)/2$. For a rank-at-most-one contraction T on B , put*

$$d = \|\text{Tr}_B F\|_1, \quad \nu = \|\text{Tr}_B[(T \otimes I)F]\|_1.$$

Then

$$\nu \leq \frac{1+d}{2}, \quad \nu^2 \leq \frac{1}{2} + \frac{1}{2}d^2. \quad (26)$$

The operator T and the contracted output need not be Hermitian.

Proof. We prove a stronger centered inequality for every Hermitian F . Write $K = \text{Tr}_B[(T \otimes I)F]$, $G = \text{Tr}_B F$, $t = \|T\|$, and $\tau = \text{Tr} T$. Then

$$\|2K - \tau G\|_1 \leq t \|F\|_1. \quad (27)$$

The case $T = 0$ is immediate. Otherwise factor $T = |u\rangle\langle v|$, so $\|u\|\|v\| = t$. For a unit vector z on B , put $P_z = |z\rangle\langle z|$ and $q = \langle z|T|z\rangle$. Two applications of Cauchy–Schwarz give

$$|q| + |q - \tau| \leq \|P_z u\| \|P_z v\| + \|(I - P_z)u\| \|(I - P_z)v\| \leq t.$$

Thus $S = 2T - \tau I$ satisfies $|\langle z|S|z\rangle| \leq t$ for every unit z . For any unitary V on A , diagonalizing V shows that $\text{Re}(S \otimes V)$ is a direct sum of blocks $\text{Re}(e^{i\theta} S)$, each of operator norm at most t . Here $\text{Re} X = (X + X^*)/2$. Hermiticity of F and trace-norm duality therefore give

$$\|2K - \tau G\|_1 = \max_{V \text{ unitary}} \text{Re} \text{Tr}[(S \otimes V)F] \leq t \|F\|_1.$$

No Hermiticity of K or T is used. Since $\|F\|_1 \leq 1$ and $|\tau| \leq t \leq 1$, the triangle inequality yields $2\nu \leq 1 + d$. Finally, for $0 \leq d \leq 1$,

$$1 - \nu^2 \geq \frac{(1-d)(3+d)}{4} \geq \frac{1}{2}(1-d^2).$$

This proves (26). $\square$

The coefficient $1/2$ in the last deficit bound is optimal. For $d \in [0, 1]$, take two-qubit states $\rho_0 = |00\rangle\langle 00|$, $\rho_1 = d|01\rangle\langle 01| + (1-d)|10\rangle\langle 10|$, with first qubit B and second qubit A , and take $T = |0\rangle\langle 0|$ on B . Then

$$F_A = \frac{d}{2}(|0\rangle\langle 0| - |1\rangle\langle 1|), \quad K = \frac{1}{2}(|0\rangle\langle 0| - d|1\rangle\langle 1|),$$

so $\|F_A\|_1 = d$ and $\nu = (1+d)/2$. The ratio $(1-\nu^2)/(1-d^2) = (3+d)/(4(1+d))$ tends to $1/2$.

C.2 A majorant using the original weights

Theorem C.3 (Generic post-insertion retention). *Let ρ_0, ρ_1 be density operators on the tensor product of finitely many finite-dimensional ports indexed by $\mathcal{J}$, with distinguished owner $0 \in \mathcal{J}$. Write*

$$F = (\rho_0 - \rho_1)/2, \quad \sigma = (\rho_0 + \rho_1)/2, \quad h_j = 1 - \lambda_{\max}(\sigma_j),$$

*where σ_j is the marginal on port j . Identify a subset of port indices with the union of their tensor factors, and write $F_A = \text{Tr}_{\mathcal{J} \setminus A} F$ for $A \subseteq \mathcal{J}$. Assume $h_0 = 1/2$ and $\sum_j h_j \leq C * s$. Insert any rank-at-most-one contraction T on a union B of whole ports, and set $K = \text{Tr}_B[(T \otimes I)F]$, $\nu = \|K\|_1$. On the remaining ports $\mathcal{R} = \mathcal{J} \setminus B$, define*

$$f(A) = \min \left\{ \nu^2, 1 - \frac{1}{2} \left(1 - 2 \sum_{j \in A} h_j \right) * +^2 \right\}.$$

Then, for every $A \subseteq \mathcal{R}$,

$$\|\text{Tr} * \mathcal{R} \setminus AK\|_1^2 \leq f(A) \leq \nu^2 \leq 1, \quad |f(A \triangle \{j\}) - f(A)| \leq 2h_j.$$

The oscillation bound is for every $j \in \mathcal{R}$. Independent fair retention of the remaining ports gives

$$\mathbb{E}f \leq 1 - \frac{1}{4}(1 - 2/e)^2 < \frac{63}{64}.$$

This includes an inserted owner and empty inserted or remaining registers, with no normalization after insertion. An empty register is one-dimensional, so its operators are scalars.

Proof. Choose a top eigenvector z_j of each original marginal. For $s(A) = \sum_{j \in A} h_j$ the commuting-projection union bound gives

$$\left\langle \bigotimes_{j \in A} z_j \left| \sigma_A \right| \bigotimes_{j \in A} z_j \right\rangle \geq 1 - s(A).$$

Lemma C.1 therefore bounds $\|F_A\|_1^2 \leq 1 - (1 - 2s(A))_+^2$. First tracing all ports outside $B \cup A$ and then applying Lemma C.2 proves the un-clipped majorant. Equation (14), with the traced-register contraction equal to the identity, also bounds the retained norm by ν , even though K need not be Hermitian. Thus clipping at ν^2 preserves the majorant.

The function $g(s) = 1 - (1 - 2s)_+^2/2$ is 2-Lipschitz on $[0, \infty)$. Clipping at a fixed number is nonexpansive; toggling port j changes $s(A)$ by exactly h_j . This proves the original-weight oscillation bound.

Let $\phi(s) = (1 - 2s)_+^2$, which is nonnegative, decreasing, and convex. If the owner remains, it is erased with probability one half. Conditional on this erasure, the mean of $s(A)$ is at most $\frac{1}{2} \sum_{j \neq 0} h_j \leq 1/e$. Jensen's inequality gives $\mathbb{E}\phi(s(A)) \geq \frac{1}{2}(1 - 2/e)^2$. If the owner was inserted, the unconditional mean already obeys $\mathbb{E}s(A) \leq 1/e$, giving the stronger bound $(1 - 2/e)^2$. Since $e > 8/3$, the resulting deficit in $\mathbb{E}g$ is greater than $1/64$. Clipping can only decrease the mean.

For empty B , T is a scalar and the preceding contraction lemma applies in a one-dimensional space. For empty A , the original F_A is the scalar zero. A zero T gives $f = 0$. No inverse of an overlap, a marginal eigenvalue, or a retained norm is used. $\square$

C.3 An independent replacement for the insertion calculation

In the exact heavy-gate expansion of Theorem B.3, replace Lemma B.2 by Theorem C.3. The original one-reflection sources satisfy its hypotheses by part (i) of Lemma 4.1. It applies to each fixed joint insertion, including the empty insertion; the resource continues to be bounded jointly as before.

For the sum of these majorants, the new bounds are

$$\mathbb{E}F_* \leq \frac{63r}{64}, \quad c_g = 2 \sum_i h_{ig}, \quad \sum_g c_g^2 \leq 4C * srw.$$

Use threshold $127r/128$, whose gap above the mean is at least $r/128$. Bounded differences and AM–GM give

$$|\mathcal{T}_{S,\ell}| \leq (127/128)^{r/2} + e^{-r/(32768C*sw)} \leq 2e^{-c'_0 r/w}, \quad c'_0 = (32768C * s)^{-1}.$$

Here $w \geq 1$ and $\frac{1}{2} \log(128/127) > c'_0$. The dense-set penalty (23) is unchanged: it uses the original joint marginal, not this new majorant. The sparse/dense summation proof consequently applies with c'_0 in place of c_0 . This gives an exponential source bound without the compressed-matrix calculation in Lemma B.2 or the uninserted retained-norm formula (10).

For a fully explicit specialization take

$$\eta = 2^{-24}, \quad \delta = 2^{-36}, \quad w = 4096.$$

Then $c'_0 > 1/40960$, $A_0 < 21 \cdot 2^{-24}$, $c'_0 - A_0 > 23/10^6$, and $c_\delta > 1/60$. The discarded-load condition follows from $2\delta \log(1/\delta) < 2^{-30} < \eta/2$; the dense-sum condition follows from $21 - 4096/120 < -1$. Thus $a = \eta/w = 2^{-36}$, and

$$K_0 < 4 + \frac{5}{(27/10)(23/10^6)} < 100000.$$

The logarithmic first-layer cutoff is $L = 78$ and $\kappa = 2^{-36}/78 > 2^{-43}$. The same output pullback therefore yields the alternative explicit estimate

$$|\beta(C)| \leq 600000 \exp(-2^{-43}n).$$

This weaker numerical constant gives the same exponential and size-independent conclusion through a different retention proof.

C.4 Sharper constants from the original monotone majorant

There is also a particularly short replacement for the post-insertion matrix calculation if one retains the uninserted source geometry from Section 4. Denote its majorant by f_0 . It is nondecreasing under inclusion, has mean at most $3/4$ under fair retention of all ports, and has oscillations at most $4h_j$.

Fix any inserted union B and a rank-at-most-one contraction T on it. Put $K = \text{Tr}_B[(T \otimes I)F]$, and, for a remaining port set A , write K_A for its marginal on A and F_A for the marginal of the original source on A . For a fair retained subset A of the remaining ports, couple it with independent fair choices on B . Monotonicity gives

$$\mathbb{E}_A f_0(A) \leq \mathbb{E}_{A,A_B} f_0(A \cup A_B) \leq 3/4.$$

This remains true when B contains the owner or all ports. For each A , Lemma C.2, applied after tracing the other remaining ports, gives

$$\|K_A\|_1^2 \leq \frac{1 + \|F_A\|_1^2}{2} \leq \frac{1 + f_0(A)}{2}.$$

Consequently

$$f_B(A) = \min\{\|K\|_1^2, (1 + f_0(A))/2\}$$

is a post-insertion majorant with mean at most $7/8$ and original-weight oscillations at most $2h_j$. Its cap is the fixed norm of the whole K , not a retention-dependent normalization.

In the heavy-set expansion use threshold $15r/16$. The mean gap is $r/16$ and $\sum_g c_g^2 \leq 4C * srw$. Thus every coefficient-free branch obeys

$$|\mathcal{T}_{S,\ell}| \leq (15/16)^{r/2} + e^{-r/(512C*sw)} \leq 2e^{-c_0''r/w}, \quad c_0'' = (512C * s)^{-1}.$$

The dense-set penalty and its subsequent summation are again unchanged. Choose now

$$\eta = 2^{-18}, \quad \delta = 2^{-30}, \quad w = 2048.$$

Then $c_0'' > 1/640$, $A_0 < 17 \cdot 2^{-18}$, $c_0'' - A_0 > 149/10^5$, and $c_\delta > 1/50$. The discarded-load condition follows from $2\delta \log(1/\delta) < 2^{-24} < \eta/2$; the dense-sum condition follows from $17 - 20.48 < -1$. It follows that $a = 2^{-29}$ and

$$K_0 < 4 + \frac{5}{(27/10)(149/10^5)} < 1500.$$

The first-layer cutoff is $L = 64$ exactly, so $\kappa = 2^{-35}$. This yields the further explicit bound

$$|\beta(C)| \leq 9000 \exp(-2^{-35}n).$$

This last route uses the original uninserted retained-norm formula but none of the post-insertion compressed-matrix identities. The preceding generic-weight route avoids both retained-matrix calculations. Both routes retain the same reflection-specific source budget and joint dense-set estimate.

D A convex joint-loss estimate and a simpler dense-set sum

This appendix replaces the small-trigger cutoff in the dense-set argument by a convex sourcewise estimate. It retains the exact heavy-gate expansion of Appendix B, the monotone-majorant transfer of Section C.4, and the first-layer and output interfaces. Thus it is a refinement of that proof, not an independent proof of those shared interfaces.

Lemma D.1 (Convex joint deficit). *For one source from Section 4 and a union B of its original ports, put*

$$W(B) = \sum_{j \in B} h_j, \quad H(B) = 1 - \lambda_{\max}(\bar{\rho}_B).$$

Define

$$\Phi(0) = 0, \quad \Phi(t) = \frac{t}{4 \log(4/t)} \quad (0 < t \leq C * s).$$

Then Φ is increasing and convex on $[0, C * s]$, and $H(B) \geq \Phi(W(B))$ for every B .

Proof. Write $W = W(B)$ and $H = H(B)$. The empty union and $W = 0$ are immediate. If B contains the owner, then $H = 1/2$ and $W \leq C * s < 5/4$. Since $4/W > 16/5 > e$, we have $\Phi(W) < W/4 < 5/16 < 1/2$.

Suppose B is ancillary-only. Let P be the full ancillary trigger probability and p its probability on B . Since $W > 0$, the case $P = 0$ is excluded: all ancillary-only weights would vanish. Also $p = 1$ would make every selected port weight zero. Thus $0 < P \leq p < 1$ before any logarithm or division

below is taken. The source geometry proved in Section 4, applied also with B treated as one port, gives

$$W \leq 2P \log(1/p), \quad H \geq P(1-p), \quad P \leq p.$$

Put $x = \log(1/p) > 0$ and obtain

$$W \leq 2xe^{-x} \leq 2/e, \quad H \geq \frac{W(1-e^{-x})}{2x}. \quad (28)$$

For $0 < x \leq 1$, the decreasing function $(1-e^{-x})/x$ is at least $1-e^{-1} > 5/8$, so $H > 5W/16$. Meanwhile $\log(4/W) \geq \log(2e) > 1$, giving $\Phi(W) < W/4 < H$.

For $x \geq 1$, the first inequality in (28) implies $\log(4/W) \geq x - \log(x/2)$. Differentiation shows that $\log(x/2)/x$ has global maximum $1/(2e) < 3/16$ on $x > 0$. Therefore

$$\frac{(1-e^{-x})\log(4/W)}{x} > \frac{5}{8} \frac{13}{16} = \frac{65}{128} > \frac{1}{2}.$$

Substitution into (28) proves $H \geq W/[4\log(4/W)]$. All divisions were made only after $P, W, x > 0$ was established. $>$

Finally, writing $\ell = \log(4/t) > 0$, direct differentiation gives

$$\Phi'(t) = \frac{\ell+1}{4t^2} > 0, \quad \Phi''(t) = \frac{\ell+2}{4t\ell^3} > 0.$$

The value and right derivative at zero are both zero. This proves the asserted monotonicity and convexity on the closed interval. $\square$

D.1 Joint aggregation and exponential summation

Use the exact heavy-gate expansion with $r \geq 1$ sources and threshold $w \geq 1$. For a selected set S of k heavy gates, let W_i be the selected original port weight of source i , and let H_i be its joint deficit on the selected union. Then $0 \leq W_i \leq C * s$ and $\sum_i W_i \geq kw$. Every source receives just one joint rank-one insertion, so its whole remaining trace norm is at most $1 - H_i$. The independent resource is kept jointly and has trace norm at most one, including when it is complex. Consequently Jensen's inequality and Lemma D.1 give

$$|\mathcal{T}_{S,\ell}| \leq \prod_i (1 - H_i) \leq \exp \left[-r\Phi \left(\frac{kw}{r} \right) \right] = \exp \left[-\frac{kw}{4\log(4r/(kw))} \right] \quad (k > 0). \quad (29)$$

Here $kw/r \leq C * s < 4$, so the denominator is positive. For $k = 0$ the middle expression equals one by continuity. No low-trigger source or selected weight is discarded.

For the light estimate, apply the bounded-differences exponential moment directly to the majorants of Section C.4. They satisfy $D_i^2 \leq f_i \leq 1$, $\mathbb{E}F_* \leq 7r/8$, and $\sum_g c_g^2 \leq 4C * srw$, where $F_* = \sum_i f_i$. For $0 < t \leq 1/2$, the inequality $\sqrt{f} \leq e^{-t(1-f)}$ holds on $[0, 1]$ by $\log f \leq f - 1$, with $f = 0$ handled directly. The exponential-moment bound (19), applied to the independent vertex coins, gives

$$|\mathcal{T}_{S,\ell}| \leq \mathbb{E} \prod_i D_i \leq e^{-tr} \mathbb{E} e^{tF_*} \leq \exp \left[-\frac{tr}{8} + \frac{t^2 C * srw}{2} \right].$$

This does not factor an expectation over sources; only vertex coins are independent. Taking $t = 1/(8C * sw) < 1/2$ yields

$$|\mathcal{T}_{S,\ell}| \leq e^{-c_0 r/w}, \quad c_0 = (128C * s)^{-1}. \quad (30)$$

The same moment bound includes constant majorants and vanishing oscillations without an exceptional-event argument. Choose $0 < \eta < C * s$ such that $A_0 = \eta \log(8eC * s/\eta) < c_0$, and put $b = c_0 - A_0 > 0$. Choose $w \geq 1$ so that

$$\log(8eC * s/\eta) - c_\eta w \leq -1, \quad c_\eta = \frac{1}{4 \log(4/\eta)}.$$

With $u = r/w$, there are at most $C * su$ heavy gates. The total absolute correction-type coefficient for a selected set of size k is 8^k .

For $1 \leq k \leq \eta u$, the binomial bound is

$$\binom{H}{k} 8^k \leq \exp\left(k \log \frac{8eC * su}{k}\right) \leq e^{A_0 u}.$$

Including $k = 0$ and at most $C * su + 1$ possible sizes, the sparse part is bounded by $(C * su + 1)e^{-bu}$. For $k > \eta u$, equation (29) gives $|\mathcal{T}_{S,\ell}| \leq e^{-c_\eta k w}$, and hence the contribution of size k is at most

$$\left(\frac{8eC * s}{\eta} e^{-c_\eta w}\right)^k \leq e^{-k}.$$

The dense sum is at most $e^{-\eta u}/(1 - e^{-1})$. These estimates include empty summation ranges and the case of no heavy gates. Thus the source theorem holds with

$$a = \frac{\min\{b/2, \eta\}}{w}, \quad K_0 = 1 + \frac{2C * s}{eb} + \frac{1}{1 - e^{-1}}. \quad (31)$$

Indeed, use $\sup_{u \geq 0} u e^{-bu/2} = 2/(eb)$ to absorb the sparse prefactor. The case $r = 0$ is bounded by one directly and is covered by $K_0 > 1$.

D.2 An explicit refined rate

Corollary D.2. *Every circuit in the model of Section 2 satisfies*

$$|\beta(C)| \leq 1800 \exp(-2^{-27} n).$$

Proof. Take $\eta = 2^{-12}$ and $w = 512$. Using $27/10 < e < 3$, $\log 2 < 7/10$, and $C * s < 5/4$, we have

$$c_0 > \frac{1}{160}, \quad \log(8eC * s/\eta) < 12, \quad b > \frac{1}{160} - 12 \cdot 2^{-12} > \frac{33}{10000}.$$

In particular $b/2 > \eta$. Also $c_\eta = 1/(56 \log 2) > 5/196$, so the dense-sum exponent is less than $12 - 640/49 = -52/49 < -1$. Equation (31) gives

$$a = 2^{-21}, \quad K_0 < 3 + \frac{5/2}{(27/10)(33/10000)} < 300.$$

The exact first-layer generating-function argument in Theorem B.4 now uses

$$L = 2 \log_2(8/a) = 48, \quad \kappa = a/L = 2^{-21}/48 > 2^{-27}.$$

Here $a < (\log 2)/2$, so that argument does not further reduce the source rate. The one-output pullback contributes coefficient mass at most six, proving the displayed bound. $\square$

The numerical constants are conservative. Two changes improve them: the convex dense-set estimate retains every source, and the exponential-moment light estimate strengthens the preceding threshold-and-tail estimate. Neither change assumes independent sampled source values or multiplies marginal penalties within one source.

Use of AI: The author prepared an 8-page draft (without using AI) with outlines of previous QAC lower bounds and several high-level proof strategies to extend them to exact parity and bounded-error parity computation using QAC circuits with depths 3, 4 and unbounded. The author then fed this draft to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) and prompted them to work out the details of the proofs and identify missing gaps. The author fixed those missing gaps and iteratively verified and expanded the proofs using AI tools. The author is solely responsible for the correctness of the proofs.

References

- [1] Johan Håstad. *Almost optimal lower bounds for small depth circuits*. STOC 1986, pages 6–20. doi:10.1145/12130.12132.
- [2] Nathan Linial, Yishay Mansour, and Noam Nisan. *Constant depth circuits, Fourier transform, and learnability*. Journal of the ACM 40(3):607–620, 1993. doi:10.1145/174130.174138.
- [3] Cristopher Moore. *Quantum circuits: fanout, parity, and counting*. arXiv:quant-ph/9903046, 1999.
- [4] Frederic Green, Steven Homer, Cristopher Moore, and Christopher Pollett. *Counting, fanout, and the complexity of quantum ACC*. Quantum Information and Computation 2(1):35–65, 2002. arXiv:quant-ph/0106017.
- [5] Peter Høyer and Robert Špalek. *Quantum fan-out is powerful*. Theory of Computing 1(5):81–103, 2005. doi:10.4086/toc.2005.v001a005.
- [6] Maosen Fang, Stephen Fenner, Frederic Green, Steven Homer, and Yong Zhang. *Quantum lower bounds for fanout*. Quantum Information and Computation 6(1):46–57, 2006. arXiv:quant-ph/0312208.
- [7] Debajyoti Bera. *A lower bound method for quantum circuits*. Information Processing Letters 111(15):723–726, 2011. doi:10.1016/j.ipl.2011.05.002.
- [8] Yasuhiro Takahashi and Seiichiro Tani. *Collapse of the hierarchy of constant-depth exact quantum circuits*. Computational Complexity 25(4):849–881, 2016. arXiv:1112.6063.
- [9] Daniel Padé, Stephen Fenner, Daniel Grier, and Thomas Thierauf. *Depth-2 QAC circuits cannot simulate quantum parity*. arXiv:2005.12169, 2020.
- [10] Gregory Rosenthal. *Bounds on the QAC⁰ complexity of approximating parity*. ITCS 2021, LIPIcs 185, 32:1–32:20. doi:10.4230/LIPIcs.ITCS.2021.32. Full version: arXiv:2008.07470v3.
- [11] Shivam Nadimpalli, Natalie Parham, Francisca Vasconcelos, and Henry Yuen. *On the Pauli spectrum of QAC⁰*. STOC 2024, pages 1498–1506. doi:10.1145/3618260.3649662. Full version: arXiv:2311.09631v4.

- [12] Joseph Slote. *Parity vs. AC^0 with simple quantum preprocessing*. ITCS 2024, LIPIcs 287, 92:1–92:21. doi:10.4230/LIPIcs.ITCS.2024.92.
- [13] Anurag Anshu, Yangjing Dong, Fengning Ou, and Penghui Yao. *On the computational power of QAC^0 with barely superlinear ancillae*. STOC 2025. arXiv:2410.06499.
- [14] Yangjing Dong, Fengning Ou, and Penghui Yao. *Linear-size QAC^0 channels: learning, testing and hardness*. arXiv:2510.00593v2, 8 November 2025.
- [15] Stephen Fenner, Daniel Grier, Daniel Padé, and Thomas Thierauf. *Tight bounds on depth-2 QAC-circuits computing parity*. arXiv:2504.06433, 2025.
- [16] Daniel Grier and Jackson Morris. *Quantum threshold is powerful*. CCC 2025. arXiv:2411.04953.
- [17] Malvika Raj Joshi, Avishay Tal, Francisca Vasconcelos, and John Wright. *Improved lower bounds for QAC^0* . STOC 2026, pages 2199–2209. doi:10.1145/3798129.3800922. Full version: arXiv:2512.14643v4, 25 August 2026.
- [18] Daniel Grier, Jackson Morris, and Kewen Wu. *QAC^0 contains TC^0 (with many copies of the input)*. arXiv:2601.03243, 2026.
- [19] Lucas Gretta, Meghal Gupta, and Malvika Raj Joshi. *Parity $\notin QAC^0 \iff QAC^0$ is Fourier-concentrated*. arXiv:2604.02793, 2026.
- [20] Yangjing Dong, Fengning Ou, and Penghui Yao. *On the computational complexity of geometrically local QAC^0 circuits*. arXiv:2604.07178, 2026.
- [21] Boyan Xu and Lvzhou Li. *Fanout complexity of symmetric Boolean functions in QAC^0* . arXiv:2609.05153, 4 September 2026.
- [22] Colin McDiarmid. *On the method of bounded differences*. In Surveys in Combinatorics, 1989, London Mathematical Society Lecture Note Series 141, pages 148–188. Cambridge University Press, 1989. doi:10.1017/CBO9781107359949.008.
- [23] Wassily Hoeffding. *Probability inequalities for sums of bounded random variables*. Journal of the American Statistical Association 58(301):13–30, 1963. doi:10.1080/01621459.1963.10500830.